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PROBLEMS OF STATISTICAL ESTIMATION AND CAUSAL INFERENCE IN TIME-SERIES REGRESSION MODELS

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Thus far in the development of quantitative social research there are relatively few examples of dynamic, time-series models of sociopolitical processes. Cross-sectional analyses are predominant in the literature. However, sociologists and political scientists are increasingly likely to utilize time-series data in their empirical research as the advantages of

the historical-dynamic perspective become more widely appreciated, and the scope and quality of longitudinal data improve. Hence it may be useful at this point to survey the problems of statistical estimation and causal inference in the time-series context. This chapter focuses on the implications of employing the classical linear regression model in the presence of autocorrelated disturbances—a common situation in time-series analysis that has important consequences for estimation and inference.

The first section of the essay reviews the classic linear regression model, which is among the most widely applied tools of quantitative social science. The following section examines the consequences of time-dependent disturbances and develops the theoretical solution—generalized least squares (GLS). Next some empirically common and mathematically tractable time-dependence models are considered, namely, autoregressive and moving-average processes. This is followed by a discussion of how the observed residuals can be analyzed to determine the process generating the disturbance interdependence, and an evaluation of what modifications in the original equation and estimation procedure are therefore appropriate. Finally these issues are considered in the context of truly dynamic models, that is, models that incorporate lagged endogenous variables on the right-hand side of the equation.

CLASSICAL LINEAR REGRESSION MODEL

Before dealing with the principal topics of this chapter, it is useful to review some of the key features of the classical linear regression model, which has become a familiar analytic tool for most quantitative social scientists. In the classical model we have[1]

$$Y = XB + U \tag{1}$$

$$E(U) = 0 \tag{2}$$

$$E(UU') = \sigma_u^2 I \tag{3}$$

$$X \text{ is fixed and has rank } (K + 1) < T \tag{4}$$

where Y is a $T \times 1$ vector of observations on the dependent or endogenous variable; X is a $T \times (1 + K)$ matrix of observations on the K independent or exogenous variables and the intercept vector (of 1s); B is a

[1] The results to follow could be shown without great difficulty to hold for the case of stochastic regressors distributed independently of U. To keep the exposition simple, I have retained the classical assumption of fixed X. Readers familiar with these well-known derivations are encouraged to pass over this section, perhaps after glancing at the key results.

$(1 + K) \times 1$ vector of coefficients to be estimated; U is a $T \times 1$ vector of disturbances; and E denotes the "expected value of," such that $E(U) = \Sigma u_t \cdot \text{prob}(u_t) = \text{mean of } U$.

The first assumption of the classical linear regression model specifies that each observation on the endogenous variable Y can be expressed as a linear function of the exogenous variables X plus the disturbance U. In the more familiar scalar notation this is expressed simply as $Y_t = \beta_0 + \sum_k \beta_k X_{tk} + u_t$. The second property of the model implies that the disturbances have no systematic components and therefore each has zero expectation. The third assumption says that the expected value of u_t^2 is σ_u^2 for all t (constant variance-homoscedasticity) and that the covariance of u_t with $u_{t\pm\theta}$ is zero whenever $\theta \neq 0$ (no autocorrelation). The final property of the classical model specifies that the X_{tk} are fixed in repeated sampling, uncorrelated with any omitted variables, and thus independent of the disturbances u_t.

Given sample data on Y and X and the prior (theoretical) assumptions of Equations (1) to (4), the researcher seeks to estimate the model

$$Y = X\hat{B} + \hat{U} \tag{5}$$

such that the sum of the squared errors $\hat{U}'\hat{U}$ is minimized. Since by definition $\hat{U}$ is expressed

$$\hat{U} = (Y - X\hat{B}) \tag{6}$$

ordinary least-squares estimation involves minimizing:

$$\begin{aligned} \hat{U}'\hat{U} &= (Y - X\hat{B})'(Y - X\hat{B}) \\ &= Y'Y - Y'X\hat{B} - (X\hat{B})'Y + (X\hat{B})'X\hat{B} \\ &= Y'Y - 2\hat{B}'X'Y + \hat{B}'X'X\hat{B} \end{aligned} \tag{7}$$

Solving for the parameters $\hat{B}$ so that $\hat{U}'\hat{U}$ is minimized requires differentiating Equation (7) with respect to $\hat{B}$:

$$\begin{aligned} \partial\hat{U}'\hat{U}/\partial\hat{B} &= 0 - 2X'Y + 2X'X\hat{B} \\ &= -X'Y + X'X\hat{B} \end{aligned} \tag{8}$$

By setting Equation (8) equal to zero and rearranging terms to isolate $\hat{B}$, we obtain the ordinary least-squares (OLS) estimator:

$$\begin{aligned} -X'Y + X'X\hat{B} &= 0 \\ X'X\hat{B} &= X'Y \\ (X'X)^{-1}X'X\hat{B} &= (X'X)^{-1}X'Y \\ \hat{B} &= (X'X)^{-1}X'Y \end{aligned} \tag{9}$$

In the case of simple bivariate regression the OLS estimator in Equation (9) has the familiar algebraic expression:

$$\hat{\beta} = \frac{\sum_t (X_t - \bar{X})(Y_t - \bar{Y})}{\sum_t (X_t - \bar{X})^2} \tag{10}$$

A few additional preliminary results need to be established by way of background to the main issues that follow. First it must be demonstrated that the OLS estimator $\hat{B}$ is unbiased; that is, on the average it hits the target parameter(s) B.[2] This is easily shown by substituting the expression for Y in Equation (1) into the expression for $\hat{B}$ in Equation (9):

$$\begin{aligned} \hat{B} &= (X'X)^{-1}X'(XB + U) \\ &= (X'X)^{-1}X'XB + (X'X)^{-1}X'U \\ &= B + (X'X)^{-1}X'U \end{aligned} \tag{11}$$

The unbiasedness of $\hat{B}$ is proved by taking the expected value of Equation (11):

$$\begin{aligned} E(\hat{B}) &= B + E[(X'X)^{-1}X'U] \\ &= B + [(X'X)^{-1}X'E(U)] \qquad \text{by Eq. (4)} \\ E(\hat{B}) &= B \qquad \text{by Eq. (2)} \end{aligned} \tag{12}$$

Finally the variance of $\hat{B}$ in the classical model must be generated because great parts of the ensuing discussion evaluate the precision and significance of parameter estimates when disturbances are autocorrelated. The variance-covariance matrix of the OLS estimator is defined as

$$\text{var}(\hat{B}) = E[(\hat{B} - B)(\hat{B} - B)'] \tag{13}$$

It is convenient to use the expression for $\hat{B}$ in Equation (11) and to rewrite Equation (13) as

$$\begin{aligned} \text{var}(\hat{B}) &= E[B + (X'X)^{-1}X'U - B][B + (X'X)^{-1}X'U - B]' \\ &= E[(X'X)^{-1}X'U][(X'X)^{-1}X'U]' \\ &= E[(X'X)^{-1}X'U\,U'X(X'X)^{-1}] \\ &= [(X'X)^{-1}X'E(UU')\;X(X'X)^{-1}] \qquad \text{by Eq. (4)} \\ &= (X'X)^{-1}X'(\sigma_u^2 I)X(X'X)^{-1} \qquad \text{by Eq. (3)} \\ \text{var}(\hat{B}) &= \sigma_u^2(X'X)^{-1} \qquad \text{(since } \sigma_u^2 \text{ is a scalar quantity and the identity matrix may be suppressed)} \end{aligned} \tag{14}$$

In the more familiar bivariate case, the variance of the OLS estimator may be expressed:

$$\text{var}(\hat{\beta}) = \sigma_u^2 / \sum_t (X_t - \bar{X})^2 \tag{15}$$

[2] More formally, an estimator is unbiased if $E(\hat{B}) = B$.

It is important to remember that least-squares regression computer programs (implicitly) assume the classical model of Equations (1) to (4) and hence routinely output the estimate of B in Equation (9), whose variance is given by (the diagonal elements of) Equation (14). The sample estimates of these equations generate the t-statistic commonly used to evaluate the precision and significance of individual regression coefficients:

$$t = \hat{B}/\sqrt{\text{var}(\hat{B})} \tag{16}$$

AUTOCORRELATED DISTURBANCES AND GENERALIZED LEAST-SQUARES ESTIMATION

The Consequences of Autocorrelation

What are the consequences of serially dependent disturbances for statistical estimation, hypothesis-testing, and causal inference? The classical model is clearly no longer appropriate. In particular, assumption (3) must be revised as follows:

$$E(UU') = \sigma_u^2\Omega \tag{17}$$

where Ω is a $T \times T$, symmetric, positive definite matrix.

The Ω specification in the revised, generalized linear regression model allows for both heteroscedasticity (nonconstant diagonal elements) and autocorrelation (nonzero off-diagonal elements). However, we are principally concerned with problems of time-series estimation and inference, and heteroscedasticity is commonly a cross-sectional problem. For our purposes, then, the Ω matrix is considered to have 1s in the diagonal and autocorrelation parameters in the off-diagonal cells. Hence in scalar notation Equation (17) implies:

$$\begin{aligned} E(u_t u_{t\pm\theta}) &= \sigma_u^2 \qquad \text{for } \theta = 0 \text{ (homoscedasticity)} \\ &= \gamma_\theta \qquad \text{for } \theta \neq 0 \text{ (autocovariance)} \end{aligned} \tag{18}$$

where γ_θ is the lag θ autocovariance.

Perhaps the first point to be made concerning the impact of autocorrelated disturbances is that OLS estimates remain unbiased. This important result is worth showing explicitly. Recall from Equation (11) that the OLS estimator of B may be expressed:

$$\hat{B} = B + (X'X)^{-1}X'U \tag{19}$$

Since in the revised, autocorrelation model X remains fixed or $E(X'U)$ remains zero, the OLS estimator still has expectation B:

$$\begin{aligned} E(\hat{B}) &= B + E[(X'X)^{-1}X'U] \\ &= B + [(X'X)^{-1}X'E(U)] \\ E(\hat{B}) &= B \end{aligned} \tag{20}$$

Hence it is possible to estimate a regression model in the conventional (OLS) manner without danger of bias even if the disturbances are serially correlated.[3]

However, the variance of $\hat{B}$ in the presence of autocorrelated disturbances is no longer that of the classical model in Equation (14), but is

$$\begin{aligned} \operatorname{var}(\hat{B}) &= E[(\hat{B} - B)(\hat{B} - B)'] \qquad \text{[see Eq. (14)]} \\ &= E[(X'X)^{-1}X'(UU')X(X'X)^{-1}] \\ &= E[(X'X)^{-1}X'(\sigma_u^2\Omega)X(X'X)^{-1}] \qquad \text{by Eq. (17)} \\ \operatorname{var}(\hat{B}) &= \sigma_u^2(X'X)^{-1}X'\Omega X(X'X)^{-1} \end{aligned} \tag{21}$$

Thus, when disturbances are interdependent, which frequently is true in time-series models, OLS regression yields biased estimates of the coefficient variances. Since the bias is generally negative, the estimated variances and standard errors understate, perhaps very seriously, the true variances and standard errors.[4] This produces inflated t-ratios, a false sense of confidence in the precision of the parameter estimates, and often leads to spurious attributions of significance to independent variables. Moreover the OLS estimate of the disturbance variance σ_u^2 is also biased, and since the bias is typically negative, R^2 as well as t- and F-statistics tend to be exaggerated.

Again this result is straightforwardly demonstrated. Note that the true disturbances are never observed directly but must be derived from the fitted model and hence are "filtered" through the Xs. Thus the residuals $\hat{u}_t$ are generated:

$$\begin{aligned} \hat{U} &= Y - X\hat{B} \\ &= Y - X(X'X)^{-1}X'Y \qquad \text{by Eq. (9)} \\ &= (XB + U) - X(X'X)^{-1}X'(XB + U) \qquad \text{by Eq. (1)} \\ &= (XB + U) - X(X'X)^{-1}X'XB - X(X'X)^{-1}X'U \\ &= (XB + U) - XB - X(X'X)^{-1}X'U \\ &= U - X(X'X)^{-1}X'U \\ &= [I_T - X(X'X)^{-1}X']U \\ &= MU \end{aligned} \tag{22}$$

where $M = [I_T - X(X'X)^{-1}X']$.

Equation (22) establishes that the residuals of the classic model are a linear function of the unknown disturbances. The sum of the

[3] As we shall see later in this chapter, this result does not obtain in dynamic models where lagged endogenous variables ($Y_{t-\theta}$) appear on the right-hand side of the equation, because then the assumption that regressors and disturbances are uncorrelated is no longer tenable.

[4] The sign and magnitude of the bias hinge on the mechanism generating the serial dependence and on the autocorrelation of the Xs. I return to this later in this chapter.

squared residuals—the quantity minimized by least-squares regression—can therefore be expressed:

$$\begin{aligned} \hat{U}'\hat{U} &= U'M'MU \\ &= U'M^2U \\ &= U'MU \qquad \text{(since } M \text{ is symmetric and idempotent)}^5 \\ &= U'[I_T - X(X'X)^{-1}X']U \qquad \text{by Eq. (22)} \end{aligned} \tag{23}$$

The expected value of Equation (23) yields the classical estimator of the disturbance sum of squares in terms of the true disturbance variance σ_u^2:

$$\begin{aligned} E(\hat{U}'\hat{U}) &= E(U'MU) \\ &= E\ \mathrm{tr}(U'MU) \qquad \text{(since } U'MU \text{ is scalar and therefore equal to its trace)} \\ &= E\ \mathrm{tr}(MUU') \qquad \text{(since tr } AB = \text{tr } BA) \\ &= \sigma_u^2\ \mathrm{tr}\ M \\ &= \sigma_u^2\ \mathrm{tr}[I_T - X(X'X)^{-1}X'] \qquad \text{by Eqs. (22) and (23)} \\ &= \sigma_u^2\ \mathrm{tr}(I_T) - \mathrm{tr}[(X'X)(X'X)^{-1}] \\ &= \sigma_u^2\ \mathrm{tr}(I_T) - \mathrm{tr}(I_{K+1}) \\ &= \sigma_u^2(T - K - 1) \end{aligned} \tag{24}$$

where K denotes the number of exogenous variables and tr denotes trace—the sum of the diagonal elements of a matrix. Thus an unbiased sample estimate of the disturbance variance in the classical case is given by

$$\hat{\sigma}_u^2 = \hat{U}'\hat{U}/(T - K - 1) \tag{25}$$

In scalar algebra this is expressed as

$$\hat{\sigma}_u^2 = \sum_t \hat{u}_t^2/(T - K - 1) \tag{26}$$

When the disturbances are autocorrelated, however, the expectation of $\hat{U}'\hat{U}$ is no longer σ_u^2 tr M, but rather

$$\begin{aligned} E(\hat{U}'\hat{U}) &= \mathrm{tr}(MUU') \qquad \text{by Eq. (24)} \\ &= \sigma_u^2\ \mathrm{tr}(M\Omega) \qquad \text{by Eq. (17)} \\ &= \sigma_u^2\ \mathrm{tr}[\Omega - X(X'X)^{-1}X'\Omega] \\ &= \sigma_u^2\ \mathrm{tr}\ \Omega - \mathrm{tr}(X'X)^{-1}X'\Omega X \\ &= \sigma_u^2\ T - \mathrm{tr}[(X'X)^{-1}X'\Omega X] \qquad \text{by the specification in Eq. (17) that } \Omega \text{ is } T \times T \text{ with 1s in the diagonal} \end{aligned} \tag{27}$$

[5] A symmetric matrix is a square matrix that is not changed by transposition, such that $M' = M$ and $m_{ij} = m_{ji}$; $i \neq j$; $ij = 1, 2, \ldots, T$. A symmetric, idempotent matrix is one that is not changed upon multiplication by itself; that is, $M' = M$ and $M^2 = M$. It is easily verified that M in (22) and (23) is symmetric and idempotent by transposing and by squaring it.

Hence the classical OLS estimator of the disturbance variance is biased to the extent that $\mathrm{tr}[(X'X)^{-1}X'\Omega X]$ differs from $\mathrm{tr}[(X'X)^{-1}X'X]$. Furthermore this bias is negative (toward zero) whenever positive autocorrelation predominates in regressors and disturbances,[6] which is generally the case for socioeconomic and political time series.

This result has two implications that are of interest here. First the $\mathrm{var}(\hat{B})$ is biased not only because $(X'X)^{-1}X'\Omega X(X'X)^{-1} \neq (X'X)^{-1}$ [see (14) and (21)], but also because $E(\hat{\sigma}_u^2) \neq \sigma_u^2$. More importantly, this means that if an equation is estimated via ordinary least squares when regressors and disturbances are (positively) autocorrelated, the researcher will obtain a spurious underestimate of the error variance and an inflation of the R^2. The model will appear to provide a much better fit to the empirical data than is actually the case.[7]

Nevertheless OLS regression in the presence of serially correlated disturbances is not necessarily disastrous—especially when the functional form of a model is not in question because of well-established theory, prior empirical results, and so on. After all, the parameter estimates are unbiased. More problematic and more typical, however, is the situation where the researcher analyzes many equations in the process of evaluating competing hypotheses and equally plausible alternative functional forms. Given the characteristic collinearity of independent variables in such studies, the erroneous selection of variables or entire equations because of differential bias in t- and F-statistics can seriously impair the causal inference-model building process. In time-series analyses of this sort, it is clear that autocorrelation is no longer a comparatively minor problem of estimation precision: if the sequential development of complex, multivariate models is grounded on biased and unreliable decision rules, errors of inference may cumulate and far exceed those that arise in a single analysis of a single equation.

The Theory of Generalized Least Squares

If the researcher has prior information about Ω, that is, knowledge of the mechanism generating the disturbance time dependence,[8] the difficulties outlined above can be avoided by applying the known Ω matrix to Equations (21) and (24). However, the sampling variances of estimates secured in this way are needlessly large compared with those

[6] See Goldberger (1964, chap. 5.4) and Malinvaud (1970, chap 13.4).

[7] Since time-series models typically have high R^2's in any case, most of the literature focuses on coefficient variances and t-statistics rather than on R^2's and (overall) F-statistics.

[8] How such information can be obtained empirically from the observed residuals is taken up later in this chapter.

rendered by Aitken's generalized least squares (GLS).[9] Consider again the generalized linear regression model:

$$Y = XB + U \tag{28}$$

$$E(U) = 0 \tag{29}$$

$$E(UU') = \sigma_u^2 \Omega \tag{30}$$

$$X \text{ is fixed and has rank } (K + 1) < T \tag{31}$$

where all terms are as previously defined.

Aitken's generalization of the Gauss-Markov least-squares theorem established that the best linear unbiased estimator of the true parameter vector B in the model of Equations (28) to (31) is the GLS estimator $\hat{B}^*$:

$$\hat{B}^* = (X'\Omega^{-1}X)^{-1}X'\Omega^{-1}Y \tag{32}$$

It is readily shown that $\hat{B}^*$ is unbiased:

$$\begin{aligned} E(\hat{B}^*) &= E[(X'\Omega^{-1}X)^{-1}X'\Omega^{-1}(XB + U)] \\ &= E[(X'\Omega^{-1}X)^{-1}X'\Omega^{-1}XB + (X'\Omega^{-1}X)^{-1}X'\Omega^{-1}U] \\ &= B + E[(X'\Omega^{-1}X)^{-1}X'\Omega^{-1}U] \\ &= B + [(X'\Omega^{-1}X)^{-1}X'\Omega^{-1}E(U)] \qquad \text{(since } X \text{ and } \Omega \text{ are fixed)} \\ E(\hat{B}^*) &= B \end{aligned} \tag{33}$$

Aitken demonstrated that the GLS estimator has minimum variance in the class of linear unbiased estimators. The variance of $\hat{B}^*$ is given by

$$\begin{aligned} &\text{var}(\hat{B}^*) \\ &\quad = E[(\hat{B}^* - B)(\hat{B}^* - B)'] \\ &\quad = E\{[B + (X'\Omega^{-1}X)^{-1}X'\Omega^{-1}U - B][B + (X'\Omega^{-1}X)^{-1}X'\Omega^{-1}U - B]'\} \\ &\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad \text{by Eq.(33)} \\ &\quad = E[(X'\Omega^{-1}X)^{-1}X'\Omega^{-1}(UU')\Omega^{-1}X(X'\Omega^{-1}X)^{-1}] \\ &\quad = [(X'\Omega^{-1}X)^{-1}X'\Omega^{-1}\sigma_u^2\Omega\Omega^{-1}X(X'\Omega^{-1}X)^{-1}] \qquad \text{by Eq. (30)} \\ &\quad = \sigma_u^2(X'\Omega^{-1}X)^{-1} \end{aligned} \tag{34}$$

Unlike the classical OLS estimator, generalized least squares yields an unbiased estimate of the error variance σ_u^2 when disturbances are autocorrelated. This is demonstrated by operations analogous to Equations (22) to (25). The residuals in the generalized model, $\hat{u}_t^*$, are defined:

[9] Aitken (1935).

$$
\begin{aligned}
\hat{U}^* &= Y - X\hat{B}^* \\
&= Y - X(X'\Omega^{-1}X)^{-1}X'\Omega^{-1}Y \qquad \text{by Eq. (32)} \\
&= (XB + U) - X(X'\Omega^{-1}X)^{-1}X'\Omega^{-1}(XB + U) \qquad \text{by Eq. (28)} \\
&= (XB + U) - X(X'\Omega^{-1}X)^{-1}X'\Omega^{-1}XB - X(X'\Omega^{-1}X)^{-1}X'\Omega^{-1}U \\
&= XB + U - XB - X(X'\Omega^{-1}X)^{-1}X'\Omega^{-1}U \qquad (35) \\
&= U - X(X'\Omega^{-1}X)^{-1}X'\Omega^{-1}U \\
&= [I_T - X(X'\Omega^{-1}X)^{-1}X'\Omega^{-1}]U \\
&= M^*U
\end{aligned}
$$

where $M^* = [I_T - X(X'\Omega^{-1}X)^{-1}X'\Omega^{-1}]$.

It can be shown that GLS minimizes a quadratic form in the residual vector $\hat{U}^*$ with Ω^{-1} as matrix; that is,

$$
\begin{aligned}
(Y - X\hat{B}^*)'\Omega^{-1}(Y - X\hat{B}^*) &= \hat{U}^{*\prime}\Omega^{-1}\hat{U}^* \\
&= (M^*U)'\Omega^{-1}(M^*U) \qquad \text{by Eq. (35)} \\
&= U'M^{*\prime}\Omega^{-1}M^*U \qquad (36)
\end{aligned}
$$

is minimized.[10] Direct calculation shows that $M^{*\prime}\Omega^{-1}M^* = \Omega^{-1}M^*$. It follows that the expected value of the sum of squared residuals from (Eq. 36) gives the GLS estimator of the disturbance sum of squares:

$$
\begin{aligned}
E(\hat{U}^{*\prime}\Omega^{-1}\hat{U}^*) &= E(U'\Omega^{-1}M^*U) \\
&= E\,\mathrm{tr}(U'\Omega^{-1}M^*U) \qquad \text{(since } U'\Omega^{-1}M^*U \text{ is scalar and therefore equal to its trace)} \\
&= E\,\mathrm{tr}(M^*UU'\Omega^{-1}) \qquad \text{(since tr } AB = \text{tr } BA) \\
&= \mathrm{tr}(M^*\sigma_u^2\Omega\Omega^{-1}) \qquad \text{by Eq. (30)} \\
&= \sigma_u^2\,\mathrm{tr}\,M^* \qquad (37) \\
&= \sigma_u^2\,\mathrm{tr}[I_T - X(X'\Omega^{-1}X)^{-1}X'\Omega^{-1}] \qquad \text{by Eq. (35)} \\
&= \sigma_u^2\,\mathrm{tr}(I_T) - \mathrm{tr}[(X'\Omega^{-1}X)^{-1}X'\Omega^{-1}X] \\
&= \sigma_u^2\,\mathrm{tr}(I_T) - \mathrm{tr}(I_{K+1}) \\
&= \sigma_u^2(T - K - 1)
\end{aligned}
$$

Hence the unbiased GLS estimate of the disturbance variance is obtained by

$$\sigma_u^2 = \hat{U}^{*\prime}\Omega^{-1}\hat{U}^*/(T - K - 1) \qquad (38)$$

The results in this section establish that generalized least squares provides what is in theory the optimum solution to the problems created by autocorrelated disturbances. However, GLS presumes prior knowledge of the disturbance variance-covariance matrix Ω, and of course this is not in general available. It is important, therefore, to consider some common mechanisms that generate serial dependence in the disturbances of regression models so that the character of Ω can more readily be deduced from sample data.

[10] See Theil (1971, p. 239ff.).

SOME MODELS FOR TIME-DEPENDENT DISTURBANCES

We noted above that to employ GLS and thereby secure efficient, linear, unbiased estimates of B and $\text{var}(\hat{B})$ in models with serially correlated errors, the investigator must have information about the variance-covariance matrix Ω. However, it is not possible to determine the $[T \times (T + 1)]/2$ distinct elements of Ω in a "barefoot," empirical fashion. Therefore it is necessary to ascertain (from the residuals of preliminary OLS regressions) the process generating the time-dependence and thereby to characterize Ω in terms of a smaller number of parameters. This section considers—again at the theoretical level—some of the most typical processes: autoregressive and moving-average models. The following section then explores how the observed residuals can be "squeezed" to reveal the dependence process that is operative in a particular regression model in order that sample estimation of Ω can be undertaken.

First-Order Autoregressive Processes

The time-dependence model that has by far received the most attention in the econometric literature is the first-order autoregressive process [AR(1)].[11] Here each disturbance u_t depends only on its own previous value (the Markov property) and a random, "white noise" component. The basic model is as follows:

$$Y_t = \beta_0 + \sum_k \beta_k X_{tk} + u_t \qquad \text{(equation to be estimated)} \tag{39}$$

$$u_t = \phi_1 u_{t-1} + \nu_t \qquad \text{(disturbance time-dependence process)} \tag{40}$$

$$-1 < \phi_1 < 1 \qquad \text{(stationarity condition)}^{12} \tag{41}$$

[11] Virtually all econometrics texts, and most experimental studies, focus on this model. An important exception is the literature on distributed lags, which I treat briefly later on. The attention given the AR(1) process probably stems from its mathematical-statistical tractability or perhaps from a widespread assumption that AR(1) disturbance processes prevail empirically. However, since alternative models are seldom entertained, the empirical regularity of AR(1) remains an open question. Apparently it is also widely believed in the econometric world that reliance on a "truncated" GLS estimator, that is, applying AR(1) even though a higher-order process may be operative, is superior to OLS in any case. This would seem to be false prima facie by simply staring at the Ω matrices that correspond to alternative time-dependence models. More convincing is a recent study by Engle (1973) which demonstrates that OLS dominates truncated GLS in some important cases. More analytic (and experimental) work along these lines is surely needed.

[12] Stationarity implies that the u_t vary about a fixed mean (in this case zero) with variance and autocovariances independent of time.

$$\begin{aligned} E(u_t) = E(\nu_t) = E(u_{t-\theta}\nu_t) &= 0 && \left.\begin{matrix}\theta > 0\\ \theta = 0\\ \theta \neq 0\end{matrix}\right\}\text{for all } t \\ E(\nu_t\nu_{t\pm\theta}) &= \sigma_\nu^2 \\ &= 0 \end{aligned} \tag{42}$$

The variance of u_t in the AR(1) process is derived as follows:

$$\begin{aligned} \sigma_u^2 &= E(u_t^2) \\ &= E[(\phi_1 u_{t-1} + \nu_t)(\phi_1 u_{t-1} + \nu_t)] \\ &= \phi_1^2 E(u_{t-1}^2) + 2\phi_1 E(u_{t-1}\nu_t) + E(\nu_t^2) \end{aligned} \tag{43}$$

By assumption (41) that the u_t follow a stationary stochastic process and therefore have identical variance for all t, and by assumption (42) that the lagged u_t are independent of the random ν_t components, the result in Eq. (43) becomes

$$\begin{aligned} \sigma_u^2 &= \phi_1^2\sigma_u^2 + \sigma_\nu^2 \\ \sigma_u^2 - \phi_1^2\sigma_u^2 &= \sigma_\nu^2 \\ \sigma_u^2(1 - \phi_u^2) &= \sigma_\nu^2 \\ \sigma_u^2 &= \sigma_\nu^2/(1 - \phi_1^2) = \gamma_0 \end{aligned} \tag{44}$$

where γ_0 denotes the variance of u. The autocovariance and autocorrelation function for the AR(1) model (denoted γ_θ and ρ_θ respectively) are similarly generated. Autocovariances are derived by multiplying through the expression for u_t [Eq. (40)] by lag $u_{t-\theta}$ and taking expectations. Hence the autocovariance of u_t and u_{t-1} is obtained:

$$\begin{aligned} \gamma_1 &= E[u_{t-1}(\phi_1 u_{t-1} + \nu_t)] \\ &= \phi_1 E(u_{t-1}^2) + E(u_{t-1}\nu_t) \\ &= \phi_1\sigma_u^2 \qquad \text{by Eqs. (41) and (42)} \end{aligned} \tag{45}$$

Parallel operations give the autocovariance of u_t and u_{t-2}:

$$\begin{aligned} \gamma_2 &= E[u_{t-2}(\phi_1 u_{t-1} + \nu_t)] \\ &= \phi_1 E(u_{t-2}u_{t-1}) + E(u_{t-2}\nu_t) \\ &= \phi_1 E(u_{t-2}u_{t-1}) \qquad \text{by Eq. (42)} \\ &= \phi_1\gamma_1 \qquad \text{by Eq. (41)} \\ &= \phi_1^2\sigma_u^2 \qquad \text{by Eq. (45)} \end{aligned} \tag{46}$$

Successive operations of the same sort would show the general autocovariance function for a first-order autoregressive model to be

$$\begin{aligned} \gamma_\theta &= \phi_1^{|\theta|}\sigma_u^2 \\ &= \phi_1^{|\theta|}\gamma_0 \end{aligned} \tag{47}$$

It is now easy to derive the general autocorrelation function for

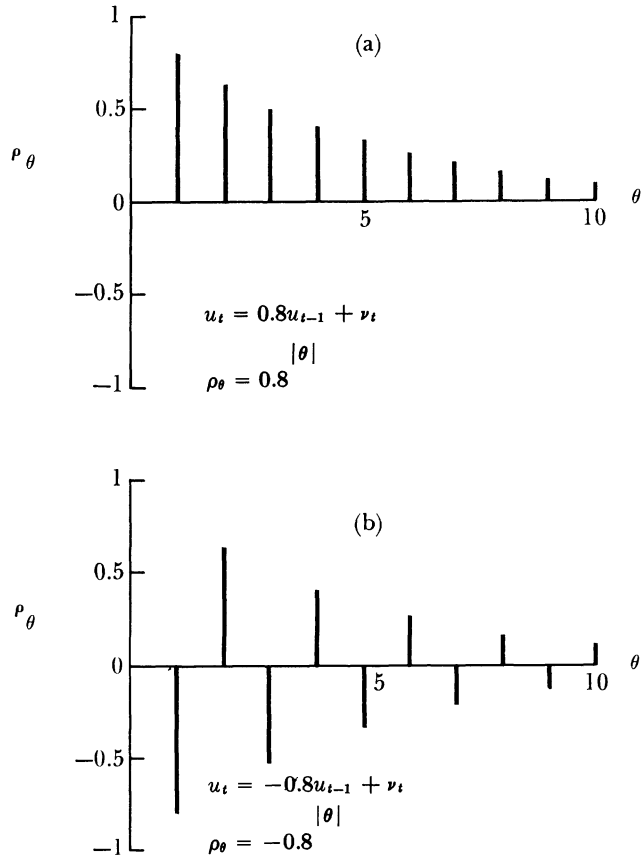

Figure 1. Theoretical Correlograms for AR(1) Disturbance Processes.

AR(1) processes. The (auto)correlation of u_t and $u_{t\pm\theta}$ is conventionally defined:

$$\rho_\theta = \frac{\operatorname{cov}(u_t u_{t\pm\theta})}{\sqrt{\operatorname{var} u_t}\sqrt{\operatorname{var} u_{t\pm\theta}}} = \rho_{-\theta} \tag{48}$$
$$= \gamma_\theta/\sigma_u^2 = \gamma_\theta/\gamma_0 \qquad \text{by Eq. (41)}$$

It now follows directly from Equation (47) that the lag autocorrelation function of the u_t is[13]

$$\rho_\theta = \phi_1^{|\theta|} \tag{49}$$

[13] Note that in the AR(1) model the autoregressive parameter ϕ_1 equals the lag 1 autocorrelation coefficient ρ_1. Hence the autocorrelation function is sometimes written $\rho_\theta = \rho_1^{|\theta|}$. It is also easily verified that $\rho_\theta = \rho_{-\theta}$ and that $\gamma_\theta = \gamma_{-\theta}$.

Equation (49) is an important result because it describes (theoretically) the autocorrelation function when the disturbance is in fact generated by a first-order autoregressive mechanism. The correlograms in Figure 1, where θ (time lags) form the abscissa and ρ_θ (autocorrelation coefficients) form the ordinate, illustrate the AR(1) autocorrelation function for two hypothetical cases. Notice that when there is positive autoregression ($\phi_1 > 0$) the ρ_θ damp off smoothly and exponentially, whereas negative autoregression ($\phi_1 < 0$) produces oscillatory, exponential decay of successive ρ_θ. As we shall see in the next section, correlogram analysis plays a central role in identifying the time-dependence process characterizing the disturbances of a particular regression model.[14]

Before considering several other dependence processes to be introduced in this section, it is useful to pursue two diversions that build on the foregoing results and motivate (somewhat belatedly) the rather abstract discussion of autocorrelated disturbances and GLS estimation developed previously.

The Consequences of First-Order Autoregression

Simple regression models in which the disturbance and regressor-follow a stationary, first-order autoregressive process have been thoroughly investigated by econometricians,[15] and it is worth relating some of the results to the theoretical material presented earlier. Consider, then, a bivariate regression model where both regressor and disturbance are stationary stochastic AR(1) processes:

$$y_t = \beta x_t + u_t \tag{50}$$

$$u_t = \phi_1 u_{t-1} + \nu_t \tag{51}$$

$$x_t = \lambda x_{t-1} + \omega_t \tag{52}$$

$$\begin{aligned} -1 < \phi_1 < 1 \\ -1 < \lambda < 1 \end{aligned} \tag{53}$$

where all variables are mean deviates.

Equation (21) established that the variance of the parameter vector $\hat{B}$ when disturbances are interdependent is

$$\operatorname{var}(\hat{B}) = \sigma_u^2 (X'X)^{-1} X'\Omega X (X'X)^{-1} \tag{54}$$

[14] The topics developed here and in following sections by covariance analysis in the time domain may also be approached via spectral analysis in the frequency domain. See Dhrymes (1970, chap. 9–12), Durbin (1969), Fishman (1969), Granger (1964), Hannan (1960, 1970), Jenkins (1961), and Jenkins and Watts (1968).

[15] See especially Rao and Griliches (1969).

In the simple example of Equations (50) to (53) this general formula is closely approximated in large samples by[16]

$$\operatorname{var}(\hat{\beta}) = \frac{\sigma_u^2}{\sum_t (x_t^2)} \left[\frac{1 + \phi_1\lambda}{1 - \phi_1\lambda}\right] \tag{55}$$

Ordinary least squares, however, routinely produces an estimate of the variance of $\hat{B}$ equal to

$$\operatorname{var}(\hat{B}) = \hat{\sigma}_u^2(X'X)^{-1} \qquad \text{[see Eq. (15)]} \tag{56}$$

In the bivariate regression model, the OLS estimate of Equation (56) becomes

$$\operatorname{var}(\hat{\beta}) = \hat{\sigma}_u^2 / \left[\sum_t (x_t^2)\right] \tag{57}$$

Comparison of the OLS estimate in Equation (57) to the true variance in Equation (55) shows clearly that the conjunction of positive or negative autocorrelation in regressor and disturbance (both are typically positive) produces a serious underestimation of $\operatorname{var}(\hat{\beta})$. For example, if $\phi_1 = \lambda = 0.8$ the bracketed ratio in Equation (55) would be

$$\frac{1 + 0.64}{1 - 0.64} = 4.56 \tag{58}$$

Hence OLS would underestimate the true variance of $\hat{\beta}$ by 456 percent and inflate the t-ratio by more than 200 percent.[17] It is easy to see, therefore, how one might erroneously conclude that a variable exerts significant causal influence if the model in question suffers from serially correlated errors and is estimated by ordinary least squares.

GLS as OLS after Transformation

This second diversion develops the specific matrix operations for the AR(1) model that are implied by the general expressions presented previously, and also illustrates how GLS estimation can be undertaken by OLS after simple transformation of variables.

Recall that our attention is confined to equations in which dis-

[16] Rao and Griliches (1969, appendix, eq. 1).

[17] Actually the inflation factor is even greater because $\hat{\sigma}_u^2$ underestimates the true error variance σ_u^2 [see Eqs. (24) and (27)]. However, this component of the bias becomes less significant as sample size gets large. Malinvaud (1970, p. 522) presents a table showing the bias factor for various combinations of ϕ_1 and λ that includes the contribution of $\hat{\sigma}_u^2$. Notice that when the independent variable is not autocorrelated the bias is small even if ϕ_1 is large.

turbances are serially correlated but not heteroscedastic. Hence:

$$E(UU') = \sigma_u^2 \Omega \tag{59}$$

such that $E(u_t u_{t\pm\theta}) = \sigma_u^2 \qquad \theta = 0$
$\qquad\qquad\qquad\quad = \gamma_\theta \qquad \theta \neq 0$

Previously the autocorrelation function of the u_t in the AR(1) model was shown to be

$$\rho_\theta = \phi_1^{|\theta|} \qquad \text{[see Eq. (49)]} \tag{60}$$

The matrix expression for the disturbance interdependence represented by Eq. (59) therefore takes the form:

$$E(UU') = \sigma_u^2 \underset{(T \times T)}{\begin{bmatrix} 1 & \phi_1 & \phi_1^2 & \phi_1^3 & \cdots & \phi_1^{T-1} \\ \phi_1 & 1 & \phi_1 & \phi_1^2 & \cdots & \phi_1^{T-2} \\ \phi_1^2 & \phi_1 & 1 & \phi_1 & \cdots & \phi_1^{T-3} \\ \vdots & & & & & \vdots \\ \phi_1^{T-1} & \phi_1^{T-2} & \phi_1^{T-3} & \phi_1^{T-4} & \cdots & 1 \end{bmatrix}} \tag{61}$$

Equation (60) specifies the character of the matrix Ω; however, GLS estimation requires Ω^{-1}. Recall:

$$\hat{B}^* = (X'\Omega^{-1}X)^{-1}X'\Omega^{-1}Y \qquad \text{[see Eq. (32)]} \tag{62}$$

and

$$\text{var}(\hat{B}^*) = \sigma_u^2(X'\Omega^{-1}X)^{-1} \qquad \text{[see Eq. (34)]} \tag{63}$$

It is readily verified that Ω^{-1} is

$$\Omega^{-1} = 1/(1-\phi_1^2) \underset{(T \times T)}{\begin{bmatrix} 1 & -\phi_1 & 0 & \cdots & 0 & 0 \\ -\phi_1 & 1+\phi_1^2 & -\phi_1 & \cdots & 0 & 0 \\ 0 & -\phi_1 & 1+\phi_1^2 & \cdots & 0 & 0 \\ \vdots & & & & & \vdots \\ 0 & 0 & 0 & \cdots & 1+\phi_1^2 & -\phi_1 \\ 0 & 0 & 0 & \cdots & -\phi_1 & 1 \end{bmatrix}} \tag{64}$$

The minimum-variance, linear-unbiased GLS estimates of B and $\text{var}(\hat{B})$ can now be secured by inserting Ω^{-1} into Eqs. (62) and (63). However, this is rather cumbersome for the nonprogramer who does not have ac-

cess to a computer package that performs matrix operations. It is in practice more convenient, therefore, to find a nonsingular transformation matrix, say A, such that:

$$\begin{aligned} A'A &= \Omega^{-1} \\ A\Omega A' &= I \end{aligned} \tag{65}$$

GLS estimation may then be achieved by applying OLS to the original model after premultiplication by A.[18] That is, if the equation to be estimated is premultiplied by A, such that $AY = AXB + AU$, then OLS applied to the transformed variables—$Y^* = AY$, $X^* = AX$, and $U^* = AU$—is equivalent to GLS.

Consider the case of the OLS estimator of B in the transformed data:

$$\begin{aligned} \hat{B}_{\text{OLS}} &= (X^{*\prime}X^*)^{-1}X^{*\prime}Y^* \qquad \text{[see Eq. (9)]} \\ &= [(AX)'AX]^{-1}(AX)'AY \\ &= [X'(A'A)X]^{-1}X'(A'A)Y \\ &= (X'\Omega^{-1}X)^{-1}X'\Omega^{-1}Y = \hat{B}^*_{\text{GLS}} \qquad \text{by Eq. (65)} \end{aligned} \tag{66}$$

The equivalence can easily be shown to hold for var($\hat{B}$) as well. Thus OLS regression of the transformed observations Y^* and X^* amounts to GLS. The disturbances of the transformed model now of course satisfy the classical assumptions:

$$\begin{aligned} E(U^*U^{*\prime}) &= E(AUU'A') \\ &= A\sigma_u^2\Omega A' \\ &= \sigma_u^2 A\Omega A' \\ &= \sigma_u^2 I \qquad \text{by Eq. (65)} \end{aligned} \tag{67}$$

If the disturbance autocorrelation is generated by a first-order autoregressive process and Ω is therefore characterized by Eq. (61), the transformation matrix A is well known to be[19]

$$A = 1/\sqrt{1-\phi_1^2}\begin{bmatrix} \sqrt{1-\phi_1^2} & 0 & 0 & \cdots & 0 & 0 \\ -\phi_1 & 1 & 0 & \cdots & 0 & 0 \\ 0 & -\phi_1 & 1 & \cdots & 0 & 0 \\ \vdots & & & & & \vdots \\ 0 & 0 & 0 & \cdots & 1 & 0 \\ 0 & 0 & 0 & \cdots & -\phi_1 & 1 \end{bmatrix} \tag{68}$$

$$(T \times T)$$

[18] A theorem in matrix algebra ensures that a matrix A with the properties of (65) exists for positive definite Ω. See Hadley (1961, chap. 7).

[19] See Theil (1971, p. 253).

Execution of the matrix multiplications $Y^* = AY$ and $X^* = AX$ confirms that the transformed variables are constructed as follows:[20]

$$\begin{aligned} Y_1^* &= \sqrt{1-\phi_1^2}\, Y_1 & t &= 1 \\ Y_t^* &= Y_t - \phi_1 Y_{t-1} & t &= 2 \ldots T \\ X_{k1}^* &= \sqrt{1-\phi_1^2}\, X_{k1} & t &= 1; k = 1 \ldots K \\ X_{kt}^* &= X_{kt} - \phi_1 X_{kt-1} & t &= 2 \ldots T; k = 1 \ldots K \end{aligned} \tag{69}$$

When Y_t^* is regressed on the X_{kt}^* it is clear from Equation (67) that the errors also are (implicitly) transformed:

$$u_t^* = u_t - \phi_1 u_{t-1} \qquad t = 2 \ldots T \tag{70}$$

Since the u_t are assumed to follow an AR(1) mechanism it is apparent that the errors of the revised model are

$$\begin{aligned} u_t^* &= \phi_1 u_{t-1} + \nu_t - \phi_1 u_{t-1} \\ u_t^* &= \nu_t \end{aligned} \tag{71}$$

Recall that the ν_t are independent random variables and so the revised model satisfies the classical assumptions.[21] Therefore ordinary least-squares regression performed on the transformed variables Y_t^* and X_{tk}^* is equivalent to generalized least squares and yields minimum-variance, linear-unbiased estimates of B and var($\hat{B}$).

Second- and Higher-Order Autoregressive Processes

Although AR(1) processes have received the most attention in the literature, there is no reason to expect a priori that autocorrelation of disturbances in time-series regression models will be generated by such a simple, albeit appealing, mechanism. It is probably accurate to say,

[20] The scalar $1/\sqrt{1-\phi_1^2}$ affects all elements of the transformed disturbance variance-covariance matrix equally and is therefore disregarded.

[21] Notice that this means the residual variance of the transformed model estimates σ_ν^2 rather than σ_u^2. If we retained the scalar $1/\sqrt{1-\phi_1^2}$ when applying the transformation matrix A to the raw data, then an estimate of σ_u^2 would be secured. That is,

$$\begin{aligned} u_t^* &= \phi_1\, u_{t-1} + \nu_t - \phi_1\, u_{t-1}/\sqrt{1-\phi_1^2} \\ u_t^* &= \nu_t/\sqrt{1-\phi_1^2} \end{aligned}$$

which has variance $\sigma_{u_t^*}^2 = \sigma_{\nu_t}^2/(1-\phi_1^2) = \sigma_u^2$ [see Eq. (44)]. Since GLS computer algorithms typically disregard such scalars the estimates of error variance and regression standard error are not comparable to the corresponding OLS statistics (see the empirical examples in the following sections). An informed discussion of the problems with commonly used GLS goodness-of-fit statistics is provided by Buse (1973). I am grateful to G. Markus for raising this issue.

however, that autoregressive processes of order higher than two are relatively uncommon—unless the data have cyclical or seasonal variability, in which case appropriate dummy variables should appear in the model.

Consider, then, a regression model where the disturbance follows a second-order autoregressive scheme (AR(2)] such that u_t depends on u_{t-1}, u_{t-2} and a random perturbation:

$$Y_t = \beta_0 + \sum_k \beta_k X_{tk} + u_t \qquad \text{(equation to be estimated)} \tag{72}$$

$$u_t = \phi_1 u_{t-1} + \phi_2 u_{t-2} + \nu_t \qquad \text{(disburbance time-dependence process)} \tag{73}$$

$$\left.\begin{aligned} \phi_2 + \phi_1 &< 1 \\ \phi_2 - \phi_1 &< 1 \\ -1 < \phi_2 &< 1 \end{aligned}\right\} \text{(stationarity conditions)} \tag{74}$$

$$\begin{aligned} E(u_t) &= E(\nu_t) = E(u_{t-\theta}\nu_t) = 0 && \theta > 0 \\ E(\nu_t \nu_{t\pm\theta}) &= \sigma_\nu^2 && \theta = 0 \\ &= 0 && \theta \neq 0 \end{aligned} \quad \text{for all } t \tag{75}$$

The variance of u_t in the AR(2) model is conveniently derived as follows:

$$\begin{aligned} \sigma_u^2 &= E(u_t)^2 \\ &= E[u_t(\phi_1 u_{t-1} + \phi_2 u_{t-2} + \nu_t)] && \text{by Eq. (73)} \\ &= \phi_1\gamma_1 + \phi_2\gamma_2 + \sigma_\nu^2 && \text{(since the only part of } u_t \text{ correlated with } \nu_t \text{ is the current perturbation } \nu_t) \end{aligned} \tag{76}$$

Dividing Eq. (76) by σ_u^2 allows the variance of the process to be expressed in terms of ϕs, ρs, and σ_ν^2:

$$\begin{aligned} \sigma_u^2/\sigma_u^2 &= \phi_1\rho_1 + \phi_2\rho_2 + \sigma_\nu^2/\sigma_u^2 && \text{by Eq. (48)} \\ 1 - \phi_1\rho_1 - \phi_2\rho_2 &= \sigma_\nu^2/\sigma_u^2 \\ \sigma_u^2 &= \sigma_\nu^2/(1 - \phi_1\rho_1 - \phi_2\rho_2) = \gamma_0 \end{aligned} \tag{77}$$

The autocovariance and autocorrelation functions are generated analogously by multiplying the equation for u_t through by $u_{t\pm\theta}$ and taking expectations. Accordingly the autocovariance of u_t and u_{t-1} is

$$\begin{aligned} \gamma_1 &= E[u_{t-1}(\phi_1 u_{t-1} + \phi_2 u_{t-2} + \nu_t)] \\ &= \phi_1 E(u_{t-1}^2) + \phi_2 E(u_{t-1}u_{t-2}) + E(u_{t-1}\nu_t) \\ &= \phi_1\sigma_u^2 + \phi_2\gamma_1 && \text{by Eqs. (74) and (75)} \end{aligned} \tag{78}$$

Similarly the autocovariance of u_t and u_{t-2} is derived:

$$\begin{aligned} \gamma_2 &= E[u_{t-2}(\phi_1 u_{t-1} + \phi_2 u_{t-2} + \nu_t)] \\ &= \phi_1 E(u_{t-2}u_{t-1}) + \phi_2 E(u_{t-2}^2) + E(u_{t-2}\nu_t) \\ &= \phi_1\gamma_1 + \phi_2\sigma_u^2 && \text{by Eqs. (74) and (75)} \end{aligned} \tag{79}$$

Recalling that γ_0 denotes σ_u^2, successive operations would show the autocovariance function of the second-order autoregressive process to be

$$\gamma_\theta = \phi_1\gamma_{\theta-1} + \phi_2\gamma_{\theta-2} \qquad \theta > 0 \tag{80}$$

The autocorrelation function for this model follows straightforwardly. Recall that ρ_θ is defined as follows:

$$\rho_\theta = \gamma_\theta/\sigma_u^2 = \gamma_\theta/\gamma_0 \tag{81}$$

Hence we need only divide the autocovariance function in Eq. (80) by γ_0 to obtain the AR(2) autocorrelation function:

$$\begin{aligned} \rho_\theta &= (\phi_1\gamma_{\theta-1} + \phi_2\gamma_{\theta-2})/\gamma_0 \\ \rho_\theta &= \phi_1\rho_{\theta-1} + \phi_2\rho_{\theta-2} \qquad \theta > 0 \end{aligned} \tag{82}$$

Again our interest centers primarily on the autocorrelation function[22] because it specifies the behavior of the ρ_θ when the disturbances in a particular equation do indeed follow an AR(2) process. Figure 2 depicts hypothetical correlograms for various combinations of the parameters ϕ_1 and ϕ_2. Note how these correlograms contrast with the autocorrelation functions of AR(1) processes illustrated in Figure 1.

Autoregressive processes of order greater than two are likely to be less common empirically, but are worth introducing briefly. The essentials involve minor extensions of previous results. Here the disturbance is generated by a pth-order autoregressive scheme:

$$u_t = \phi_1 u_{t-1} + \phi_2 u_{t-2} + \cdots + \phi_p u_{t-p} + \nu_t \tag{83}$$

The variance of u_t is easily shown to be

$$\sigma_u^2 = \frac{\sigma_\nu^2}{1 - \rho_1\phi_1 - \rho_2\phi_2 - \cdots - \rho_p\phi_p} \tag{84}$$

The autocovariance function is

$$\gamma_\theta = \phi_1\gamma_{\theta-1} + \phi_2\gamma_{\theta-2} + \cdots + \phi_p\gamma_{\theta-p} \qquad \theta > 0 \tag{85}$$

Finally the autocorrelation function of a pth-order autoregressive model is given by

$$\rho_\theta = \phi_1\rho_{\theta-1} + \phi_2\rho_{\theta-2} + \cdots + \phi_p\rho_{\theta-p} \qquad \theta > 0 \tag{86}$$

These general expressions, and in particular Eq. (86), can be used to deduce the empirical behavior of disturbances generated by an auto-

[22] Note that Eq. (82) yields $\phi_1\rho_0 + \phi_2\rho_{-1}$ as the expression for ρ_1. Since $\rho_0 = 1$ and $\rho_{-\theta} = \rho_\theta$, this becomes $\rho_1 = \phi_1 + \phi_2\rho_1$. Hence ρ_1, the starting point of the AR(2) autocorrelation function, may be written as: $\rho_1 = \phi_1/(1 - \phi_2)$.

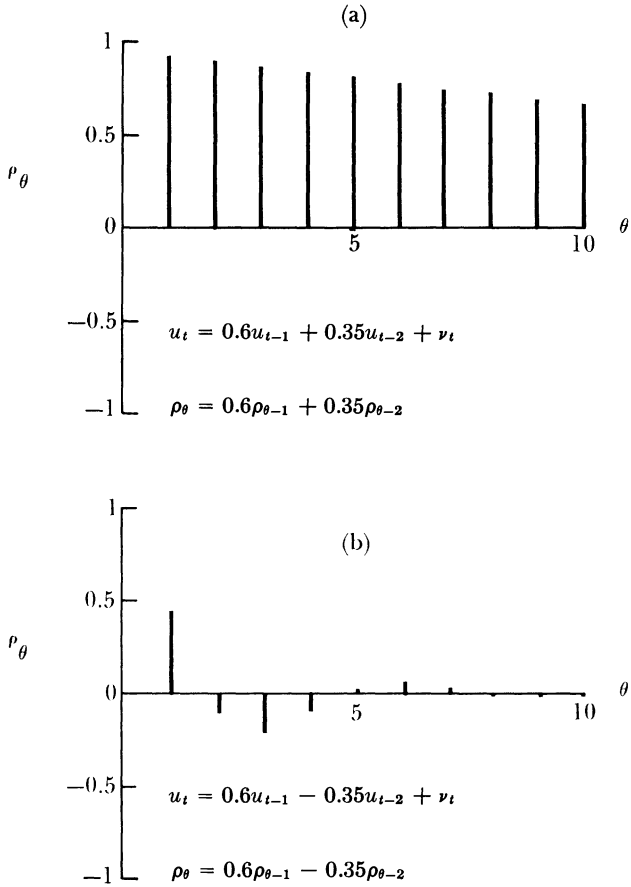

Figure 2. Theoretical Correlograms for AR(2) Disturbance Processes.

regressive process of any order. We pursue this further after considering moving average models.

First-Order Moving Average Processes

An alternative class of models for time-dependent disturbances is provided by moving average processes. In contrast to autoregressive models, disturbances generated by moving-average schemes depend only on a moving linear combination of random variables ν_t with coefficients $(1, -\phi_1, \ldots, -\phi_p)$. Hence a random shock ν_t enters the system at time t and disturbs the equilibrium level of u_t for $(p + 1)$ periods before dissipation. The autocorrelation functions and correlograms produced

by moving-average processes therefore differ sharply from those of autoregressive models.

In the first-order moving-average process [MA(1)] we have

$$Y_t = \beta_0 + \sum_k \beta_k X_{tk} + u_t \qquad \text{(equation to be estimated)} \tag{87}$$

$$u_t = \nu_t - \phi_1 \nu_{t-1} \qquad \text{(disturbance time-dependence process)} \tag{88}$$

$$-1 < \phi_1 < 1 \qquad \text{(invertibility condition)}^{23} \tag{89}$$

$$\left.\begin{aligned} E(u_t) &= E(\nu_t) = E(u_{t-\theta}\nu_t) = 0 \\ E(\nu_t\nu_{t\pm\theta}) &= \sigma_\nu^2 \qquad\qquad \theta > 0 \\ &= 0 \qquad\qquad\qquad\quad \theta \neq 0 \end{aligned}\right\} \text{for all } t \tag{90}$$

The variance of the first-order moving-average model may be derived as follows:

$$\begin{aligned} \sigma_u^2 &= E(u_t^2) \\ &= E[(\nu_t - \phi_1\nu_{t-1})(\nu_t - \phi_1\nu_{t-1})] \\ &= E(\nu_t^2) - 2\phi_1 E(\nu_{t-1}\nu_t) + \phi_1^2 E(\nu_{t-1}^2) \\ &= \sigma_\nu^2 + \phi_1^2\sigma_\nu^2 \qquad \text{by Eq. (90)} \\ \sigma_u^2 &= \sigma_\nu^2(1 + \phi_1^2) = \gamma_0 \end{aligned} \tag{91}$$

The MA(1) autocovariance function is similarly generated. Thus the autocovariance of u_t and u_{t-1} is

$$\begin{aligned} \gamma_1 &= E(u_t u_{t-1}) \\ &= E[(\nu_t - \phi_1\nu_{t-1})(\nu_{t-1} - \phi_1\nu_{t-2})] \\ &= E(\nu_t\nu_{t-1}) - \phi_1 E(\nu_t\nu_{t-2}) - \phi_1 E(\nu_{t-1}^2) + \phi_1^2 E(\nu_{t-1}\nu_{t-2}) \\ &= -\phi_1\sigma_\nu^2 \qquad \text{by Eq. (90)} \end{aligned} \tag{92}$$

Parallel operations yield autocovariances of greater lag. For example, γ_2 is obtained:

$$\begin{aligned} \gamma_2 &= E(u_t u_{t-2}) \\ &= E[(\nu_t - \phi_1\nu_{t-1})(\nu_{t-2} - \phi_1\nu_{t-3})] \\ &= E(\nu_t\nu_{t-2}) - \phi_1 E(\nu_t\nu_{t-3}) - \phi_1 E(\nu_{t-1}\nu_{t-2}) + \phi_1^2 E(\nu_{t-1}\nu_{t-3}) \\ &= 0 \qquad \text{by Eq. (90)} \end{aligned} \tag{93}$$

It should now be apparent that in first-order moving-average processes all autocovariances beyond lag 1 are zero. Thus the general autocovariance function for the MA(1) model is written:

$$\begin{aligned} \gamma_\theta &= -\phi_1\sigma_\nu^2 \qquad \theta = \pm 1 \\ &= 0 \qquad\qquad |\theta| > 1 \end{aligned} \tag{94}$$

[23] The invertibility conditions for moving-average processes parallel the stationarity conditions of autoregressive models by defining the admissible region for the parameters ϕ_1. See Box and Jenkins (1970, chap. 3.3).

Obviously the autocorrelation function of the MA(1) process, defined by $\rho_\theta = \gamma_\theta/\gamma_0$, shares the same property. Thus

$$\rho_1 = \frac{\gamma_1}{\gamma_0} = \frac{-\phi_1\sigma_\nu^2}{(1+\phi_1^2)\sigma_\nu^2} = \frac{-\phi_1}{(1+\phi_1^2)}$$

and successive lag ρ_θ are zero. Therefore the general autocorrelation functron is expressed:

$$\begin{aligned} \rho_\theta &= -\phi_1/(1+\phi_1^2) & \theta &= \pm 1 \\ &= 0 & |\theta| &> 1 \end{aligned} \tag{95}$$

The results in Eqs. (94) and (95) establish that the autocovariances and autocorrelations of the first-order moving-average process have a cutoff after lag 1. This produces a correlogram (see Figure 3) that is readily distinguished from the dampoff of the AR(1) correlogram

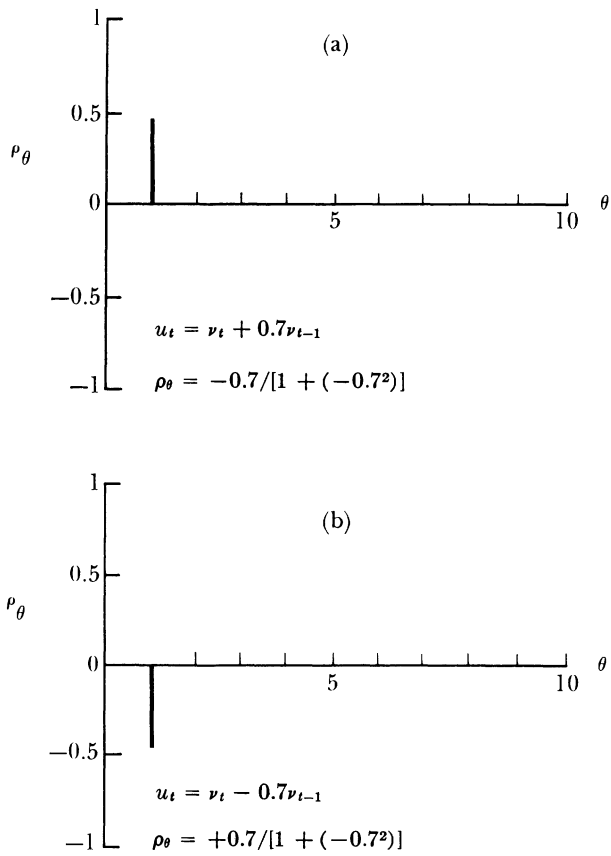

Figure 3. Theoretical correlograms for MA(1) disturbance processes.

(Figure 1)[24] and the decay pattern of AR(2)–AR(p) correlograms (Figure 2).

Second- and Higher-Order Moving-Average Processes

This discussion of second- and higher-order moving-average processes is brief since the essential results follow straightforwardly from those developed above. Consider, then, a regression model where the disturbance follows a second-order moving-average scheme [MA(2)]:

$$Y_t = \beta_0 + \sum_k \beta_k X_{tk} + u_t \qquad \text{(equation to be estimated)} \tag{96}$$

$$u_t = \nu_t - \phi_1 \nu_{t-1} - \phi_2 \nu_{t-2} \qquad \text{(disburbance time-dependence process)} \tag{97}$$

$$\begin{aligned} \phi_2 + \phi_1 &< 1 \\ \phi_2 - \phi_1 &< 1 \qquad \text{(invertibility conditions)} \\ -1 < \phi_2 &< 1 \end{aligned} \tag{98}$$

$$\begin{aligned} E(u_t) &= u_{t-\theta}\nu_t = 0 \\ E(\nu_t \nu_{t\pm\theta}) &= \sigma_\nu^2 \qquad \theta > 0 \\ &= 0 \qquad \theta \neq 0 \end{aligned} \Bigg\} \text{for all } t \tag{99}$$

The variance of the MA(2) model is

$$\begin{aligned} \sigma_u^2 &= E(u_t^2) \\ &= E[(\nu_t - \phi_1\nu_{t-1} - \phi_2\nu_{t-2})(\nu_t - \phi_1\nu_{t-1} - \phi_2\nu_{t-2})] \\ &= E(\nu_t^2) - 2\phi_1 E(\nu_t\nu_{t-1}) - 2\phi_2 E(\nu_t\nu_{t-2}) \\ &\quad + 2\phi_1\phi_2 E(\nu_{t-1}\nu_{t-2}) + \phi_1^2 E(\nu_{t-1}^2) \\ &\quad + \phi_2^2 E(\nu_{t-2}^2) \\ &= \sigma_\nu^2 + \phi_1^2\sigma_\nu^2 + \phi_2^2\sigma_\nu^2 \qquad \text{by Eq. (99)} \\ \sigma_u^2 &= \sigma_\nu^2(1 + \phi_1^2 + \phi_2^2) = \gamma_0 \end{aligned} \tag{100}$$

[24] Similarly the MA(1) Ω matrix contrasts sharply with that of the AR(1) model [see Eq. (61)]:

$$E(UU') = \sigma_u^2 \begin{bmatrix} 1 & \dfrac{-\phi_1}{1+\phi_1^2} & 0 & \cdots & 0 \\ \dfrac{-\phi_1}{1+\phi_1^2} & 1 & \dfrac{-\phi 1}{1+\phi_1^2} & \cdots & 0 \\ \vdots & & & & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix}$$

$$(T \times T)$$

The autocovariance function is derived analogously:

$$\begin{aligned}
\gamma_1 &= E(u_t u_{t-1}) \\
&= E[(\nu_t - \phi_1\nu_{t-1} - \phi_2\nu_{t-2})(\nu_{t-1} - \phi_1\nu_{t-2} - \phi_2\nu_{t-3})] \\
&= E(\nu_t\nu_{t-1}) - \phi_1 E(\nu_t\nu_{t-2}) - \phi_2 E(\nu_t\nu_{t-3}) \\
&\quad - \phi_1 E(\nu_{t-1}^2) + \phi_1^2 E(\nu_{t-1}\nu_{t-2}) \\
&\quad + \phi_1\phi_2 E(\nu_{t-1}\nu_{t-3}) - \phi_2 E(\nu_{t-2}\nu_{t-1}) \\
&\quad + \phi_1\phi_2 E(\nu_{t-2}^2) + \phi_2^2 E(\nu_{t-2}\nu_{t-3}) \\
&= (-\phi_1 + \phi_1\phi_2)\sigma_\nu^2 \qquad \text{by Eq. (99)}
\end{aligned} \tag{101}$$

$$\begin{aligned}
\gamma_2 &= E(u_t u_{t-2}) \\
&= E[(\nu_t - \phi_1\nu_{t-1} - \phi_2\nu_{t-2})(\nu_{t-2} - \phi_1\nu_{t-3} - \phi_2\nu_{t-4})] \\
&= E(\nu_t\nu_{t-2}) - \phi_1 E(\nu_t\nu_{t-3}) - \phi_2 E(\nu_t\nu_{t-4}) \\
&\quad - \phi_1 E(\nu_{t-1}\nu_{t-2}) + \phi_1^2 E(\nu_{t-1}\nu_{t-3}) + \phi_1\phi_2 E(\nu_{t-1}\nu_{t-4}) \\
&\quad - \phi_2 E(\nu_{t-2}^2) + \phi_2\phi_1 E(\nu_{t-2}\nu_{t-3}) + \phi_2^2 E(\nu_{t-2}\nu_{t-4}) \\
&= -\phi_2\sigma_\nu^2 \qquad \text{by Eq. (99)}
\end{aligned} \tag{102}$$

$$\begin{aligned}
\gamma_3 &= E(u_t u_{t-3}) \\
&= E[(\nu_t - \phi_1\nu_{t-1} - \phi_2\nu_{t-2})(\nu_{t-3} - \phi_1\nu_{t-4} - \phi_2\nu_{t-5})] \\
&= E(\nu_t\nu_{t-3}) - \phi_1 E(\nu_t\nu_{t-4}) - \phi_2 E(\nu_t\nu_{t-5}) \\
&\quad - \phi_1 E(\nu_{t-1}\nu_{t-3}) + \phi_1^2 E(\nu_{t-1}\nu_{t-4}) + \phi_1\phi_2 E(\nu_{t-1}\nu_{t-5}) \\
&\quad - \phi_2 E(\nu_{t-2}\nu_{t-3}) + \phi_2\phi_1 E(\nu_{t-2}\nu_{t-4}) + \phi_2^2 E(\nu_{t-2}\nu_{t-5}) \\
&= 0 \qquad \text{by Eq. (99)}
\end{aligned} \tag{103}$$

In generalized notation, the autocovariance function for the second-order moving-average process is therefore:

$$\begin{aligned}
\gamma_\theta &= (-\phi_\theta + \phi_1\phi_{\theta+1})\sigma_\nu^2 \qquad \theta = \pm 1, 2 \\
&= 0 \qquad \theta > \pm 2
\end{aligned} \tag{104}$$

Since $\rho_\theta = \gamma_\theta/\gamma_0$, the general autocorrelation function for the MA(2) mechanism follows directly:

$$\begin{aligned}
\rho_\theta &= \frac{(-\phi_\theta + \phi_1\phi_{\theta+1})\sigma_\nu^2}{(1 + \phi_1^2 + \phi_2^2)\sigma_\nu^2} \qquad \theta = \pm 1, 2 \\
\rho_\theta &= \frac{-\phi_\theta + \phi_1\phi_{\theta+1}}{1 + \phi_1^2 + \phi_2^2} \qquad \theta = \pm 1, 2 \\
&= 0 \qquad |\theta| > 2
\end{aligned} \tag{105}$$

Hence in the second-order moving-average process the autocovariance and autocorrelation functions are zero after lag 2. The corresponding correlograms (see Figure 4) are therefore easily differentiated from those of autoregressive processes and lower (or higher) order moving-average processes. Moreover, moving-average models of higher order [MA(p)] have analogous properties; that is, autocovariances and autocorrelations exhibit a cutoff beyond lag p.

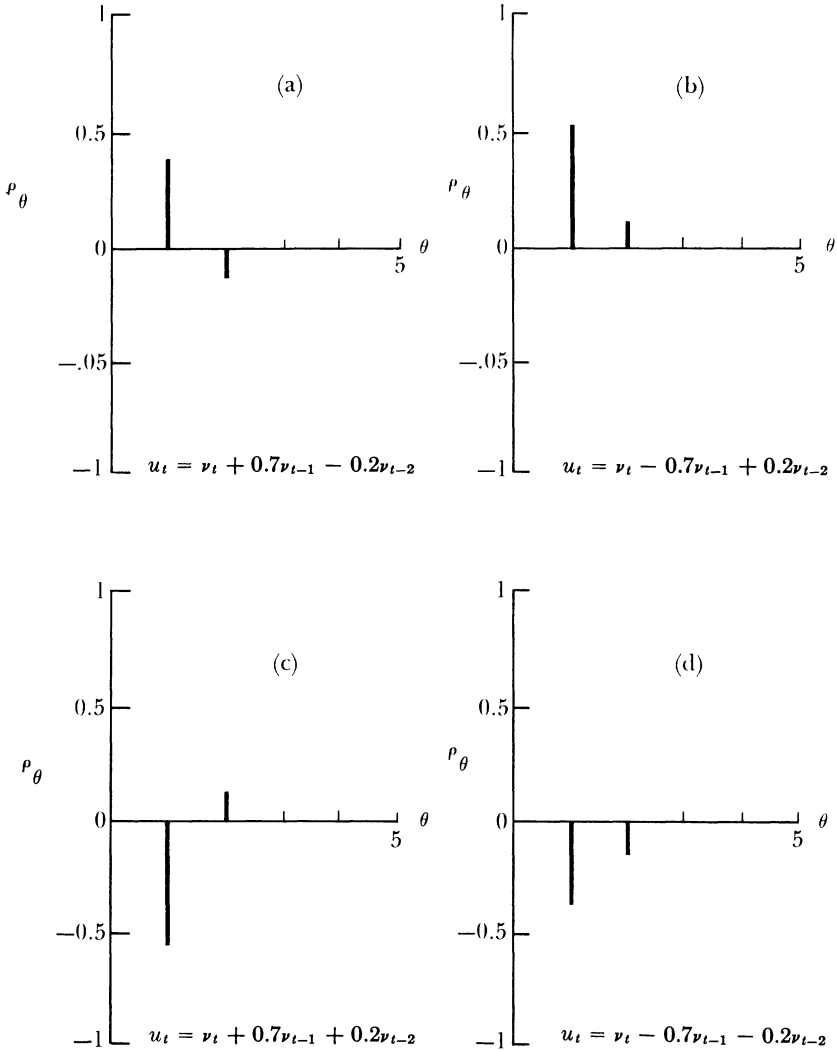

Figure 4. Theoretical correlograms for MA(2) disturbance processes.

Briefly, in the pth-order moving-average process we have

$$u_t = \nu_t - \phi_1\nu_{t-1} - \phi_2\nu_{t-2} - \cdots - \phi_p\nu_{t-p} \tag{106}$$

The u_t in this model have variance

$$\sigma_u^2 = \sigma_\nu^2(1 + \phi_1^2 + \phi_2^2 + \cdots + \phi_p^2) = \gamma_0 \tag{107}$$

and autocovariance

$$\begin{aligned} \gamma_\theta &= (-\phi_\theta + \phi_1\phi_{\theta+1} + \phi_2\phi_{\theta+2} + \cdots + \phi_{p-\theta}\phi_p)\sigma_\nu^2 && \theta = \pm 1, 2, \ldots, p \\ &= 0 && |\theta| > p \end{aligned} \tag{108}$$

Finally, and most importantly, the autocorrelation function of the MA(p) process is

$$\begin{aligned} \rho_\theta &= \frac{-\phi_\theta + \phi_1\phi_{\theta+1} + \cdots + \phi_{p-\theta}\phi_p}{1 + \phi_1^2 + \cdots + \phi_p^2} && \theta = \pm 1, 2, \ldots, p \\ &= 0 && |\theta| > p \end{aligned} \tag{109}$$

Like higher-order autoregressive models, moving-average processes of order greater than two are likely to be rare in practice, although just how rare is essentially an empirical question.

INFORMATION ABOUT Ω: CORRELOGRAM ANALYSIS AND PSEUDO-GENERALIZED LEAST SQUARES ESTIMATION

Previous results established that generalized least squares is the optimum method to estimate regression models with autocorrelated disturbances. However, GLS requires knowledge of the disturbance variance-covariance matrix Ω whose $[T \times (T + 1)]/2$ distinct elements can be ascertained only if the disturbance interdependence is generated by a known process. This issue led to our examination of some empirically common, mathematically tractable models for time-dependent disturbances. After deriving the variance, autocovariance, and autocorrelation functions of these models, we indicated that correlogram analysis is an effective way to deduce the time-dependence process characterizing the disturbances of a particular regression equation. Hence, given an equation that (the researcher suspects) suffers from serially correlated errors, the magnitude and the form of the interdependence can be determined by undertaking a preliminary OLS regression, retrieving the residuals, and generating the empirical correlogram. The empirical correlogram should then be compared to the theoretical correlograms of potentially appropriate models [AR(p), MA(p)] so that an informed specification of a process can be made. The corresponding Ω, Ω^{-1} matrices and/or the appropriate transformation matrix (earlier denoted A) are then straightforwardly estimated and (pseudo) GLS estimates secured.

Theoretical and Empirical Correlograms

The efficacy of the procedure outlined above hinges in large part on the comparison of empirical to theoretical correlograms, which in practice determines the time-dependence process that is chosen. Con-

sider first autoregressive processes. Recall that the general autocorrelation function is given by

$$\rho_\theta = \phi_1\rho_{\theta-1} + \phi_2\rho_{\theta-2} + \cdots + \phi_p\rho_{\theta-p} \qquad \theta > 0 \qquad \text{[see Eq. (86)]} \tag{110}$$

Hence in the AR(1) case, where $u_t = \phi_1 u_{t-1} + \nu_t$, we have

$$\rho_\theta = \phi_1\rho_{\theta-1} \tag{111}$$

Since $\rho_1 = \phi_1$ in this process, Equation (111) may be written

$$\rho_\theta = \phi_1^{|\theta|} \qquad \text{[see Eq. (49]} \tag{112}$$

For AR(2) models, where $u_t = \phi_1 u_{t-1} + \phi_2 u_{t-2} + \nu_t$, the autocorrela-function is

$$\rho_\theta = \phi_1\rho_{\theta-1} + \phi_2\rho_{\theta-2} \qquad \text{[see Eq. (82)]} \tag{113}$$

Let ρ_θ^* denote the theoretical autocorrelations and $\hat{\rho}_\theta$ denote the corresponding empirical autocorrelations. In the context of a particular regression equation the ρ_θ^* for the AR(1) model are therefore obtained by regressing $\hat{u}_t$ on $\hat{u}_{t-1}$ (OLS residuals) to secure a sample estimate of ϕ_1, and then calculating iteratively $\rho_2^*, \rho_3^*, \ldots, \rho_\theta^*$:

$$\rho_\theta^* = \hat{\phi}_1^{|\theta|} \tag{114}$$

The ρ_θ^* of the AR(2) process are obtained analogously. First secure sample estimates of ϕ_1 and ϕ_2 from the OLS residuals by regressing $\hat{u}_t$ on $\hat{u}_{t-1}$ and $\hat{u}_{t-2}$. The ρ_θ^* are then generated recursively:

$$\rho_\theta^* = \hat{\phi}_1\rho_{\theta-1}^* + \hat{\phi}_2\rho_{\theta-2}^* \tag{115}$$

where $\rho_1^* = \hat{\phi}_1/(1 - \hat{\phi}_2)$ (see footnote 22). Similar operations yield the theoretical autocorrelations for higher-order autoregressive models. The ρ_θ^* derived by this procedure are theoretical in the sense that empirical-sample data are used only to estimate the parameters of $\phi_1 \ldots \phi_p$. Thereafter the ρ_θ^* are determined entirely by the (theoretical) autocorrelation function of the time-dependence process being considered.

Derivation of the theoretical autocorrelations for moving-average processes are more problematic. Recall that the general MA(p) autocorrelation function is

$$\begin{aligned} \rho_\theta &= \frac{-\phi_\theta + \phi_1\phi_{\theta+1} + \cdots + \phi_{p-\theta}\phi_p}{1 + \phi_1^2 + \phi_2^2 + \cdots + \phi_p^2} && \theta = \pm 1, 2, \ldots, p \\ &= 0 && |\theta| > p \end{aligned} \tag{116}$$

Unlike the situation in the analysis of autoregressive models, the parameters ϕ_p of moving-average processes cannot be estimated by direct least

squares from the OLS residuals $\hat{u}_t$. Nor can they be derived linearly from the empirical $\hat{\rho}_\theta$ (except in the simplest case where $p = 1$). Hence the theoretical ρ_θ^* are not readily generated prior to selection of a model. We do know, however, that the theoretical autocorrelation function exhibits a cutoff where $\theta > p$ (see Figures 3 and 4), so moving-average schemes generally can be identified from the empirical correlogram without formal comparison to the theoretical autocorrelation function.[25]

The empirical autocorrelations are of course obtained simply by computing successive $\hat{\rho}_\theta$ from the OLS residuals $\hat{u}_t$. That is, $\hat{\rho}_1$ equals the correlation of $\hat{u}_t$ and $\hat{u}_{t-1}$; $\hat{\rho}_2$ equals the correlation of $\hat{u}_t$ and $\hat{u}_{t-2}$; and so on. However, the observed residuals do not perfectly reflect the theoretical disturbances, and therefore one cannot expect perfect congruence between the empirical correlogram and that of the underlying theoretical process. Residuals, and hence empirical autocorrelations, are "noisy," and the noise tends to predominate as the lag increases and the true ρ_θ cutoff or dampoff. As Box and Jenkins put it: "Moderately large estimated autocorrelations can occur after the theoretical autocorrelation function has damped out, and apparent ripples and trends can occur in the estimated [empirical] function which have no basis in the theoretical function. In employing the estimated autocorrelation function as a tool for identification [model selection], it is usually possible to be fairly sure about broad characteristics, but more subtle indications may or may not represent real effects, and two or more related models may need to be entertained and investigated further at the estimation and diagnostic checking stages of model building."[26] We must anticipate, therefore, less than perfect correspondence between empirical and theoretical autocorrelations, and as a practical rule should compute and analyze no more than $T/5$ or $T/4$ $\hat{\rho}_\theta$ (where T = sample size).

Since the noise in the empirical autocorrelations prevails as θ increases, it is useful when comparing empirical to theoretical correlograms to know whether large lag $\hat{\rho}_\theta$ are effectively zero. Following Bartlett (1946) the standard deviation of large lag $\hat{\rho}_\theta$ is closely approximated by

$$\hat{\sigma}_{\hat{\rho}_\theta} = 1/T^{1/2}\,[1 + 2(\hat{\rho}_1^2 + \hat{\rho}_2^2 + \cdots + \hat{\rho}_p^2)]^{1/2} \qquad \theta > p \qquad (117)$$

where p is the lag beyond which the process is deemed to have "died out."

For moderate sample sizes, R. L. Anderson (1942) has shown that the distribution of $\hat{\rho}_\theta$, whose corresponding population value is zero, is very nearly normal. Hence the researcher can determine whether the

[25] It is sometimes difficult, however, to distinguish moving-average processes from autoregressive models that damp off quickly. See Hannan (1960, pp. 41ff.) and the ensuing discussion in this chapter.

[26] Box and Jenkins (1970, p. 177).

empirical autocorrelation function or correlogram has in fact damped off or cut off in the fashion predicted by the theoretical process being entertained by calculating the test statistic $\hat{\rho}_\theta/\hat{\sigma}_{\hat{\rho}_\theta}$, which is distributed approximately as a unit normal deviate. The hypothesis that $\hat{\rho}_\theta = 0(\theta > p)$ is then evaluated in the conventional manner.

In addition the partial autocorrelation function may be examined. Let $\rho_{t\theta}$ denote the partial correlation between u_t and $u_{t+\theta}$ holding $u_{t+1} \ldots u_{t+\theta-1}$ fixed. If the disturbances follow an autoregressive process of order p, the partial autocorrelations (as well as the partial coefficients ϕ_p) will be nonzero for $\theta \leq p$ and zero for $\theta > p$. Hence the partial autocorrelations of a pth-order autoregressive process exhibit a cutoff beyond lag p. Quenouille (1949) has established that the $\hat{\rho}_{t\theta}$ estimated in the sample are approximately independently distributed for $\theta > p$ and have standard error: $\hat{\sigma}_{\hat{\rho}_{t\theta}} \approx 1/\sqrt{T}$. Thus the test statistic $\hat{\rho}_{t\theta}/\hat{\sigma}_{\hat{\rho}_{t\theta}}$, which is distributed as a unit normal deviate, can be used to evaluate the hypothesis that $\hat{\rho}_{t\theta} = 0$ $(\theta > p)$ and thereby aid in identifying the order of an autoregressive process (or the presence of a moving-average process) in the OLS residuals.[27]

The partial autocorrelation functions of moving-average processes are complicated and so do not lend themselves to precise evaluation in the sample. However, unlike the partial autocorrelations of autoregressive models that cut off after lag p, the partial autocorrelations of disturbances generated by a moving-average process tail off or damp off in a manner analogous to the (nonpartial) autocorrelation function of AR(p) mechanisms (see Figures 1 and 2). Hence we have an important duality that facilitates the choice of a model for time-dependent disturbances: pth-order moving-average processes have autocorrelation functions that are zero after lag p, and partial autocorrelation functions that are infinite in extent and dominated by damped exponentials and/or damped sine waves. Conversely, pth-order autoregressive processes have autocorrelation functions that are infinite in extent and consist of mixed damped exponentials and/or damped sine waves, and have partial autocorrelation functions that are zero after lag p.[28]

[27] The tests given here and in (117), however, are designed for observed variables u_t rather than residuals $\hat{u}_t$, which are computed after least-squares fitting. Hence they should be undertaken with caution and only aid in detecting the order of the process. Exact tests for specific autoregressive models are provided by Durbin and Watson (1950, 1951) and Wallis (1972). Also see footnote 34.

[28] This duality extends to other features of MA and AR processes as well. See Box and Jenkins (1970, chap. 3). Also relevant to this section are T. W. Anderson (1971, chap. 5 and 6), Kendall and Stuart (1968, chap. 47 to 50), and especially Rudra (1952).

The $\hat{\rho}_\theta$ are not only noisy but are negatively biased (in small samples) as well.[29] The negative bias poses no great difficulty when comparing empirical correlograms to the various theoretical alternatives, since the ρ_θ^* are generated with empirically derived $\hat{\phi}_p$ as starting points. Hence the sample bias of $\hat{\rho}_\theta$ should not distort identification or choice of the time-dependence model. It does become troublesome, however, when undertaking sample-based (pseudo) GLS estimation.

Pseudo-GLS Estimation

The purpose of identifying the time-dependence process followed by the disturbances is, of course, to secure GLS estimates of B, var(B), and so on. Previous sections have developed the logic of GLS generally (and theoretically) and have outlined the specifics for the AR(1) case. The point to be emphasized here is that once the dependence process has been ascertained the investigator still must rely on sample estimates of the parameters ϕ_p and can therefore derive only estimates of the variance-covariance matrix Ω, its inverse Ω^{-1}, and the transformation matrix A. Since $\hat{\phi}_p$ and $\hat{\rho}_\theta$ are biased in the sample (the magnitude of the bias depending on ρ_p, T, and the autocorrelation of the exogenous variables), $\hat{\Omega}$, $\hat{\Omega}^{-1}$, and $\hat{A}$ are biased as well. Hence we use the denotation *pseudo-GLS* when ϕ_θ, ρ_θ, Ω, and A are not known exactly.

For autoregressive processes, psudo-GLS can be achieved by OLS after transformation—and therefore without access to specialized computer programs—according to the procedures previously outlined in the context of the AR(1) model.[30] To secure small-sample regression coefficient estimates that are substantially better than those produced by OLS, however, it is important to have accurate estimates of the ϕ_p. Rao and Griliches have evaluated the merits of competing estimators of ϕ_1

[29] See Kendall (1954), Kendall and Stuart (1968, chap. 48.3), and Marriott and Pope (1954).

[30] For higher-order autoregressive models the appropriate transformations are

$$Y_1^* = \sqrt{1 - \phi_1^2 - \phi_2^2 - \cdots - \phi_p^2}\, Y_1 \qquad t = 1$$
$$Y_t^* = Y_t - \phi_1 Y_{t-1} - \phi_2 Y_{t-2} - \cdots - \phi_p Y_{t-p} \qquad t = 2\ldots T$$
$$X_{k_1}^* = \sqrt{1 - \phi_1^2 - \phi_2^2 - \cdots - \phi_p^2}\, X_1 \qquad t = 1; k = 1\ldots K$$
$$X_{kt}^* = X_{kt} - \phi_1 X_{kt-1} - \phi_2 X_{kt-2} - \cdots - \phi_p X_{kt-p} \qquad t = 2\ldots T, k = 1\ldots K$$

Notice that when GLS is undertaken in this way the intercept does not estimate β_0 but: $\beta_0(1 - \phi_1 - \phi_2 - \cdots - \phi_p)$.

Direct matrix solutions are also feasible. The appropriate Ω^{-1} matrices are given by Wise (1955, p. 155).

in the AR(1) process,[31] and Malinvaud suggests some small-sample corrections for bias.[32] In AR(2) models the estimates

$$\hat{\phi}_1 = \frac{\hat{\rho}_1(1 - \hat{\rho}_2)}{1 - \hat{\rho}_1^2}$$

and

$$\hat{\phi}_2 = \frac{\hat{\rho}_2 - \hat{\rho}_1^2}{1 - \hat{\rho}_1^2}$$

approximate fully efficient maximum-likelihood estimates and are therefore preferred to those obtained from OLS residual regressions.[33] Ideally the investigator will have access to maximum-likelihood computer programs that yield the best possible estimates of the ϕ_p and proceed directly to GLS, making transformation of variables unnecessary.

Specialized (nonlinear) programs are necessary in any case to estimate the ϕ_p of moving-average models since, as we noted previously, they are nonlinear functions of the $\hat{\rho}_\theta$. Thus, moving-average processes require rather complex operations and are not easily handled unless ap-

[31] Rao and Griliches (1969). The Monte Carlo experiments ($T = 20$) reported in this study suggest that when the disturbance is generated by an AR(1) process and $|\phi_1| \geq 0.3$, pseudo-GLS offers a considerable improvement over OLS. I find this a more appealing rule of thumb than the conventional Durbin-Watson test. See Durbin and Watson (1950, 1951). Since time-series-oriented, least-squares computer programs routinely report the Durbin-Watson d statistic it is perhaps useful to point out that an estimate of ρ_1 is easily deduced from it. Recall that

$$\hat{\rho}_1 = \frac{\hat{\gamma}_1}{\hat{\gamma}_0} = \frac{\operatorname{cov}(\hat{u}_t\, \hat{u}_{t-1})}{\operatorname{var} \hat{u}_t}$$

The Durbin-Watson d is computed:

$$d = \frac{\sum_t (\hat{u}_t - \hat{u}_{t-1})^2}{\sum_t \hat{u}_t^2}$$

Expanding the previous equation gives

$$d = \frac{\sum_t \hat{u}_t^2 - 2\sum_t \hat{u}_t\, \hat{u}_{t-1} + \sum_t \hat{u}_{t-1}^2}{\sum_t \hat{u}_t^2}$$

$$d \simeq 1 - 2\hat{\rho}_1 + 1$$
$$d \simeq 2 - 2\hat{\rho}_1$$

Hence $\hat{\rho}_1$ can be deduced as follows: $\hat{\rho}_1 \simeq 1 - d/2$

[32] Malinvaud (1970, chap. 13.4).

[33] See Box and Jenkins (1970, chap. 6.3).

propriate computer algorithms are available.[34] However, it is clear that if care is taken in specifying the form of Ω (time-dependence process), and if theoretical as well as experimental results are utilized to minimize bias in the ϕ_p, pseudo-GLS is not only asymptotically efficient but also dominates OLS in small samples, as Rao and Griliches (1969) have shown.

An empirical example is surely overdue.

Presidential Popularity from Truman to Johnson

A recent, imaginative study of presidential popularity from Truman to Johnson by John Mueller provides a useful benchmark for an illustration of some of the theoretical points developed in previous sections.[35] Mueller was interested in developing a model to explain the temporal decline in the percentage of the public approving the way an incumbent is handling his job as president—a phenomenon of considerable interest for which there are some 299 data points (via the Gallup Poll) from the beginning of the Truman administration in 1945 to the end of the Johnson administration in January 1969.

Mueller presents six equations that are based on prior theory and the preliminary analysis of literally hundreds of regression models. There is of course no way to determine how Mueller's use of conventional significance tests in the presence of autocorrelation might have affected this massive process of causal inference and model reformulation. However, each of the reported equations suffers from serious serial correlation, and undoubtedly this was also true of the unreported models examined by Mueller in preparing the published results.[36]

Let us reexamine Mueller's final model (his Equation 6) in some detail. Briefly this model specifies the dependent variable—the percent-

[34] Box and Jenkins (1970) present the appropriate algorithms and provide charts that allow one to read off estimates of the ϕ_p given values of the empirical $\hat{\rho}_\theta$. See chap. 7 and part V. It is also worth noting that after specification of Ω and estimation via pseudo-GLS, the transformed errors $\hat{\nu}_t$ should of course behave as a white-noise process. Therefore autocorrelations of the $\hat{\nu}_t$ might be inspected to ensure that the set of GLS residuals are indeed serially independent. Moreover, specification errors in the functional form linking Y_t to the X_{kt} can often be detected by evaluating the cross-correlations of $\hat{\nu}_t$ and $X_{kt-\theta}$. See Box and Jenkins (1970, chap. 11), Box and Pierce (1970), Pierce (1971a, 1971b, 1972).

[35] Mueller (1970). See also Mueller (1973, chap. 9).

[36] Mueller refers to the autocorrelation problem throughout his study and notes that the Durbin-Watson statistic [which presumes an AR(1) process] is highly significant for each equation. Although this tells us nothing about the form of the disturbance time-dependence model it does indicate that $\hat{\rho}_1$ in each equation was sizable. See footnote 31.

age of the public approving the way the incumbent is handling his job as president (% Approve)—to be a linear function of:

1. A "rally round the flag" variable (Rally)—the length of time, in years, since the last rally point. Rally points include the start of each presidential term and all international events deemed dramatic enough to give a sudden boost to the president's popularity rating.
2. An "economic slump" variable (Ecoslump)—the unemployment rate at the time the incumbent's term began subtracted from the rate at the time of the poll, but set equal to zero whenever the unemployment rate was lower at the time of the survey than it had been at the start of the incumbent's present term. (A slumping economy is expected to harm a president's popularity, but an improving economy is not anticipated to help his rating.)
3. "Dummy" or binary variables for each presidential term, which are designed to permit a term-by-term comparison of the personal, idiosyncratic appeal (or lack thereof) of individual presidents (HST1, HST2; IKE1, IKE2; JFK1; LBJ1, LBJ2).
4. A "coalition of minorities" (Mincoals) variable, term-by-term (HST1 * Mincoals, HST2 * Mincoals, . . ., LBJ2 * Mincoals)—the length of time, in years, since the incumbent was inaugurated or reelected, multiplied by the appropriate presidential term, dummy variable. It allows for the progressive alienation of former supporters at a rate (slope) that varies across individual terms.
5. "War" dummy variables designed to capture the (presumably negative) impact on presidential popularity of the Korean and Vietnamese adventures (Warkorea, Warviet).

Mueller's final equation to explain presidential popularity is therefore expressed as follows:

$$\begin{aligned}\%\ \text{Approve} = {} & \beta_0 + \beta_1\,\text{Rally} + \beta_2\,\text{Ecoslump} + \beta_3\,\text{HST2} \\ & + \beta_4\,\text{IKE1} + \beta_5\,\text{IKE2} + \beta_6\,\text{JFK1} \\ & + \beta_7\,\text{LBJ1} + \beta_8\,\text{LBJ2} + \beta_9\,\text{HST1} * \text{Mincoals} \\ & + \beta_{10}\,\text{HST2} * \text{Mincoals} + \beta_{11}\,\text{IKE1} * \text{Mincoals} \\ & + \beta_{12}\,\text{IKE2} * \text{Mincoals} + \beta_{13}\,\text{JFK1} * \text{Mincoals} \\ & + \beta_{14}\,\text{LBJ1} * \text{Mincoals} + \beta_{15}\,\text{LBJ2} * \text{Mincoals} \\ & + \beta_{16}\,\text{Warkorea} + \beta_{17}\,\text{Warviet} + u_t \qquad (118)\end{aligned}$$

The theoretical rationale of the model is summarized by Mueller as follows: It is anticipated (1) that each president will experience in each

TABLE 1
OLS Estimates for Equation (118)

Independent Variable	Parameter Estimate	Standard Error	t-Statistic
Intercept (HST1)	72.38	2.49	29.02
Rally	−6.17	1.03	−5.96
Ecoslump	−3.72	0.64	−5.79
HST2	−12.42	3.53	−3.52
IKE1	−2.41	2.98	−0.81
IKE2	−4.35	2.90	−1.50
JFK1	7.18	3.10	2.31
LBJ1	4.02	3.89	1.03
LBJ2	−1.06	3.21	−0.33
HST1*Mincoals	−8.92	1.33	−6.70
HST2*Mincoals	−2.82	1.35	−2.09
IKE1*Mincoals	2.58	0.81	3.18
IKE2*Mincoals	0.22	0.62	0.35
JFK1*Mincoals	−4.75	1.15	−4.13
LBJ1*Mincoals	2.53	8.43	0.30
LBJ2*Mincoals	−8.13	0.79	−10.33
Warkorea	−18.20	3.39	−5.37
Warviet	−0.01	2.77	−0.00

$R^2 = 0.859$ Regression standard error = 5.73 $F = 100.9$; $DF = 17{,}281$

term a general decline of popularity; (2) that this decline will be interrupted from time to time by temporary upsurges associated with international crises and similar events; (3) that the decline will be accelerated in direct relation to increases in unemployment rates over those prevailing when the president began his term, but that improvement in unemployment rates will not affect his popularity one way or the other; and (4) that the president will experience an additional loss of popularity if a war is underway.

The OLS results for Eq. (118) are reported in Table 1.[37] The residuals of this equation were retrieved and subjected to correlogram analysis in order to deduce the form, as well as initial estimates of the coefficients, of the disturbance time-dependence process. A first-order autoregressive process clearly provides the best fit to the empirical autocorrelations, although, as Figure 5 indicates, the empirical function

[37] The data for this analysis were kindly made available to me by Professor Mueller. The OLS estimates in Table 1 differ slightly from those reported in the published article because they are based on Mueller's revised sample of 299 observations versus 292 used in the original study). The OLS estimates do, however, correspond perfectly to Mueller's own revised results, which appear in Mueller (1973, chap. 9).

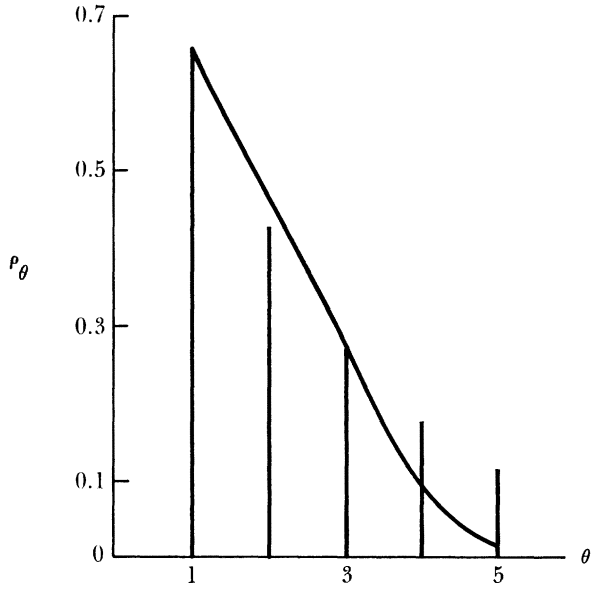

Figure 5. Empirical and theoretical correlograms for Equation (118). Curved line plots empirical autocorrelations; bar lines plot theoretical autocorrelations.

(curved line) damps off a bit more quickly than the theoretical function (bar lines). Note that the estimate of ϕ_1 used to generate the theoretical correlogram (0.650) was secured by regressing $\hat{u}_t$ on $\hat{u}_{t-1}$. The empirical autocorrelations were of course obtained simply by computing successive $\hat{\rho}_\theta$ from the OLS residuals ($\hat{u}_t$) of Eq. (118). The partial autocorrelation function also corresponded to the AR(1) pattern—that is, all $\hat{\rho}_{t\theta}$ were effectively zero.

Having established that an AR(1) process characterizes the disturbance time dependence, and having secured an initial guess of ϕ_1, Eq. (118) was reestimated by generalized least squares.[38] The GLS results appear in Table 2.

What difference does this reanalysis of Mueller's final model make? The GLS estimates should not lead to causal inferences that differ radically from those of Mueller, since we have not attempted a complete replication of his large-scale investigation but have confined

[38] The TROLL/1 econometric system was used to estimate the model. See Eisner (1972). 0.650 was provided as the initial guess of ϕ_1, which converged to the (maximum-likelihood) value of 0.823. Note that the opinion sample points are not uniformly equidistant, and so the estimate of ϕ_1 is somewhat, but not significantly, weakened. See Kendall and Stuart (1968, chap. 49.22).

TABLE 2
GLS Estimates for Equation (118)

Independent Variable	Parameter Estimate	Standard Error	t-Statistic
Intercept (HST1)	84.63	6.55	12.93
Rally	−4.92	1.14	−4.31
Ecoslump	−1.68	1.17	−1.45
HST2	−31.12	9.50	−3.28
IKE1	−20.58	7.98	−2.58
IKE2	−14.25	7.92	−1.80
JFK1	−10.22	8.19	−1.25
LBJ1	−4.52	8.89	−0.51
LBJ2	−17.71	8.22	−2.15
HST1*Mincoals	−17.02	3.01	−5.65
HST2*Mincoals	−6.18	2.45	−2.52
IKE1*Mincoals	2.81	2.15	1.31
IKE2*Mincoals	−1.15	1.80	−0.64
JFK1*Mincoals	−3.13	2.91	−1.08
LBJ1*Mincoals	−9.09	14.16	−0.64
LBJ2*Mincoals	−7.76	2.15	−3.61
Warkorea	2.49	4.37	0.57
Warviet	4.84	4.15	1.17

$R^2 = 0.551^a$ Regression standard error $= 4.10^a$ $F = 20.3^a$; $DF = 17{,}281$

[a] These GLS statistics should not be compared to the corresponding OLS statistics because they are based on the transformed disturbances AU. See the previous discussion in this chapter, especially footnote 21. Let us apply the relevant expression in footnote 21 to get some idea of the OLS understatement of the regression standard error $\sqrt{\hat{\sigma}_u^2}$ or average error of prediction in Equation (118). We have $\hat{\sigma}_u^2 = \hat{\sigma}_\nu^2/1 - \hat{\phi}_1^2$ (see footnote 21). Noting that $\sqrt{\hat{\sigma}_\nu^2} = 4.1$ (Table 2) and $\hat{\phi}_1 = 0.823$ (footnote 38), we may derive an estimate of the true OLS residual variance and regression standard error: $\hat{\sigma}_u^2 = (4.1)^2/1 - (0.823)^2 = 52.5$. The true OLS regression standard error $\sqrt{\hat{\sigma}_u^2}$ is therefore 7.25, which is considerably understated by the estimate 5.73 reported in Table 1.

the example to a single analysis of a single equation (see the previous discussion). However, the GLS estimates do indicate that substantively nontrivial modifications of Mueller's conclusions are in order. Moreover, even where the thrust of Mueller's inferences need not be altered, the exact GLS parameter values are surely more reliable than the OLS estimates whose true variances are needlessly large [see Equations (21) and (27)].

The first important difference between the GLS and OLS results involves the Ecoslump variable. The GLS estimates for this variable in Table 2 demonstrate that the significance attributed to Ecoslump in the OLS analysis is a spurious artifact of autocorrelation. Hence the rate of unemployment (as transformed by Mueller) exhibits neither a statis-

tically significant nor a theoretically important influence on presidential popularity.

The results for the presidential term dummy variables also differ markedly across estimation methods. Here the most dramatic lack of correspondence shows up in the parameter values rather than in the standard errors or *t*-statistics. The GLS estimate of 84.6 percent for the starting point of Truman's first-term popularity rating (HST1)—which as the null case in the dummy variable formulation is given by the intercept term—provides a much better fit to the empirical data than does the OLS estimate of 72.4 percent.[39] Indeed Mueller's analysis of the OLS residuals led him to make special note of the poor performance of the model in accounting for Truman's extremely high initial popularity in the aftermath of Roosevelt's death. Naturally the remaining term dummies, which are interpreted as deviations from the Truman starting point, have correspondingly larger absolute values. The Truman Mincoals or rate of alienation slopes (HSTI * Mincoals, HST2 * Mincoals) are also larger (negative) in the GLS results. Again the GLS estimates outperform the OLS results in capturing the dynamics of the enormous decline in Truman's popularity—especially during the first few years of his incumbency.

Finally the GLS results indicate that Mueller's inferences concerning the influence of undeclared war should be modified. The OLS estimates suggest that the Korean expedition had a dramatic negative impact on Truman's popularity, whereas the involvement in Vietnam had no independent effect on the mass public's approval of LBJ. Mueller attributes this rather disconcerting outcome to the collinearity of the Mincoals variables and the war dummies—and to the fact that Korea became "Truman's War" in a way that Vietnam never became "Johnson's War." The GLS parameter estimates and *t*-statistics demonstrate that the latter explanation is unnecessary and erroneous. Neither war dummy coefficient is sizable, significant, or even has the anticipated sign. A more plausible interpretation is that Mincoals variables have absorbed the antiwar effect along with other alienating factors in both cases.

INTERDEPENDENT DISTURBANCES AND DYNAMIC FORMULATIONS

Lagged Endogenous Variables and Autocorrelated Disturbances

Thus far our examination of statistical estimation and causal inference in regression equations with serially correlated disturbances has

[39] Time plots of the popularity data are provided by Mueller (1973, pp. 198–201).

been confined to static models—that is, independent variables have been exclusively exogenous and time has not appeared in an essential way. Although OLS estimation of such regression models gives rise to a number of statistical pathologies and frequently leads to erroneous causal inferences, we have seen that the coefficient estimates remain unbiased. This no longer obtains in truly dynamic models, which incorporate lagged endogenous variables on the right-hand side,[40] because it is well known that the conjunction of lagged $y_{t-\theta}$ and autocorrelated disturbances produces biased and inconsistent estimates.[41] Hence, even if the researcher has an infinite number of empirical observations at his disposal, the OLS estimates do not converge to the true parameter values.

The point is readily demonstrated for some simple cases. Consider the following bivariate equation where both regressor and disturbance follow a stationary, stochastic first-order autoregressive process:[42]

$$y_t = \beta y_{t-1} + u_t \qquad \text{(equation to be estimated)} \tag{119}$$

where y_t is a mean deviate.

$$u_t = \phi_1 u_{t-1} + \nu_t \qquad \text{(disturbance time-dependence process)} \tag{120}$$

$$\left.\begin{aligned} -1 < \beta < 1 \\ -1 < \phi_1 < 1 \end{aligned}\right\} \text{(stationarity conditions)} \tag{121}$$

$$E(u_t) = E(\nu_t) = E(u_{t-\theta}\nu_t) = E(y_{t-\theta}\nu_t) = 0 \} \text{for } \theta > 0 \text{ and all } t \tag{122}$$

If we lag Equation (119) one period and multiply through by ϕ_1 we obtain

$$\phi_1 y_{t-1} = \phi_1 \beta y_{t-2} + \phi_1 u_{t-1} \tag{123}$$

This yields a useful expression for the autoregressive component of the disturbance in the original model—$\phi_1 u_{t-1}$:

$$\phi_1 u_{t-1} = \phi_1 y_{t-1} - \phi_1 \beta y_{t-2} \tag{124}$$

Hence the original model [Equation (119)] may be written

$$\begin{aligned} y_t &= \beta y_{t-1} + \phi_1 u_{t-1} + \nu_t \\ &= \beta y_{t-1} + \phi_1 y_{t-1} - \phi_1 \beta y_{t-2} + \nu_t \\ &= (\beta + \phi_1) y_{t-1} - \phi_1 \beta y_{t-2} + \nu_t \end{aligned} \tag{125}$$

[40] For an especially good treatment of the theoretical differences between static and dynamic models in the econometric context see Christ (1966).

[41] A consistent estimator is defined: plim $\hat{\beta} = \beta$, where plim denotes "in the probability limit." An alternative but not identical formulation is "consistency in mean square": $E(\hat{\beta} - \beta)^2 \to 0$ as $N \to \infty$. Hence a consistent estimator is one whose bias and variance converge to zero as sample size becomes infinite.

[42] An exhaustive treatment of this case is provided by Griliches (1961).

It is now obvious that application of ordinary least squares to the equation as initially specified is equivalent to the classic omitted-variables situation in which the coefficient estimate of the included variable y_{t-1} picks up some influence of the excluded variable y_{t-2}.

More precisely, it is well known that if the "true" equation takes the form

$$y_t = \beta_1 x_{1t} + \beta_2 x_{2t} + \cdots + \beta_k x_{kt} + u_t$$

and the investigator mistakenly omits variables and estimates some subset of the true function, then, in the probability limit, the OLS estimates of the included parameters will be

$$\text{plim } \hat{\beta}_k = \beta_k + \sum_j \beta_j * b_{jk} \tag{126}$$

where β_k denotes the true parameter of the kth included variable; β_j denotes the true parameter of the jth omitted variable; b_{jk} denotes partial coefficients from the (multivariate) auxiliary regressions of each omitted variable on all included variables; and $*$ denotes multiplication.[43]

In the example at hand, we have $(\beta + \phi_1)$ as the true coefficient of the included variable y_{t-1}, and $-\phi_1\beta$ as the true coefficient of the omitted variable y_{t-2} [see Equation (125)]. The OLS estimate of β in the original model will therefore be

$$\begin{aligned}
\text{plim } \hat{\beta} &= (\beta + \phi_1) + [-\phi_1\beta * \text{plim } b_{y_{t-2},\, y_{t-1}}] \\
\text{plim } \hat{\beta} &= (\beta + \phi_1) + [-\phi_1\beta * \text{plim } \hat{\beta}] \\
&\quad \text{by the stationarity assumption of Eq. (121)} \qquad (127) \\
\text{plim } \hat{\beta} + (\phi_1\beta * \text{plim } \hat{\beta}) &= (\beta + \phi_1) \\
\text{plim } \hat{\beta}(1 + \phi_1\beta) &= (\beta + \phi_1) \\
\text{plim } \hat{\beta} &= (\beta + \phi_1)/(1 + \phi_1\beta)
\end{aligned}$$

It is clear, therefore, that ordinary least squares renders inconsistent parameter estimates for dynamic models with temporally dependent

[43] A matrix demonstration easily proves this result. If the "truth" is $Y = X_1B_1 + X_2B_2 + U$, and the estimated model is $Y = X_1B_1$, then the OLS estimates in the misspecified equation are given by $\hat{B}_1 = (X_1'X_1)^{-1} X_1'Y$ [see Eq. (9)]. Substituting in for Y yields $\hat{B}_1 = (X_1'X_1)^{-1} (X_1X_1) B_1 + (X_1'X_1)^{-1} (X_1'X_2) B_2 + (X_1'X_1)^{-1} X_1'U$. In the probability limit this becomes plim $\hat{B}_1 = B_1 + [AB_2]$, where $A = (X_1'X_1)^{-1} X_1'X_2$ and the bracketed expression gives the asymptotic bias. For further analysis along these lines see Griliches (1957) and Theil (1957). We focus exclusively on asymptotic properties in this section because all estimators of dynamic models exhibit small-sample bias. Only consistency or asymptotic unbiasedness can be preserved by the estimation methods to be introduced shortly.

disturbances. In simple autoregressive models of the sort in Equations (119) to (122), the asymptotic bias is

$$\text{plim } (\hat{\beta} - \beta) = \frac{\phi_1(1 - \beta^2)}{(1 + \phi_1\beta)}$$

and hence is positive (overestimating the impact of y_{t-1}) whenever ϕ_1 is positive.[44] As a result the causal influence attributed to lagged $y_{t-\theta}$ in autoregressive equations is commonly exaggerated.

Moreover the OLS residuals no longer accurately reflect the true underlying disturbances. This, of course, follows directly from the fact that the lagged $y_{t-\theta}$ have a tendency to absorb the systematic component of the disturbances. Hence the previous procedures cannot be used to ascertain the structure of the disturbance time dependence since the residual estimates of ϕ_p and ρ_θ as well as the Durbin-Watson statistic will be biased, even in unlimited samples.[45] In contrast to the comparatively minor, small-sample bias in the models of the previous section, the asymptotic bias of $\hat{\phi}_1$ in autoregressive models can be severe, especially when ϕ_1 is large and β is small. Specifically, for the model in Equations (119) to (122) it is easily shown that[46]

$$\text{plim } \hat{\phi}_1 = \frac{\beta\phi_1(\beta + \phi_1)}{(1 + \beta\phi_1)} \tag{128}$$

so that the asymptotic bias is

$$\text{plim } (\hat{\phi}_1 - \phi_1) = \frac{-\phi_1(1 - \beta^2)}{(1 + \phi_1\beta)} \tag{129}$$

Thus the bias in $\hat{\phi}_1$ is equal in size but opposite in sign to that of the regression coefficient estimate $\hat{\beta}$, and so conventional tests applied to the OLS residuals underestimate the true magnitude of the serial correlation.

The bivariate autoregressive model of Eq. (119) is useful for

[44] This result is derived as follows. Asymptotic bias is defined as $\text{plim}(\hat{\beta} - \beta)$ and we know $\text{plim}\,\hat{\beta}$ to be $(\beta + \phi_1)/(1 + \phi_1\beta)$. Thus $\text{plim}(\hat{\beta} - \beta) = [(\beta + \phi_1)/(1 + \phi_1\beta)] - \beta = [(\beta + \phi_1) - \beta(1 + \phi_1\beta)]/(1 + \phi_1\beta) = (\beta + \phi_1 - \beta - \phi_1\beta^2)/(1 + \phi_1\beta) = [\phi_1(1 - \beta^2)]/(1 + \phi_1\beta)$.

An alternative way to derive the result in Equation (127) is to notice that y_t is an AR(2) process with $\phi_1 = (\beta + \phi_1)$ and $\phi_2 = (-\phi_1\beta)$. Since β is by definition the first-order autocorrelation coefficient ρ_1, it follows that $\beta = \rho_1 = \phi_1/(1 - \phi_2) = (\beta + \phi_1)/(1 + \phi_1\beta)$ (see footnote 22). However, the omitted-variables approach we have taken here will be useful when considering more complex formulations later on.

[45] See Nerlove and Wallis (1966) on the inappropriateness of the Durbin-Watson statistic in dynamic models. Durbin (1970) has developed an autocorrelation test statistic designed especially for such situations.

[46] See Malinvaud (1970, chap. 14.5).

illustrating some basic results, but applied problems are rarely so simple. In practice, dynamic models usually incorporate one or more exogenous variables which, as Malinvaud's Monte Carlo experiments have shown, reduce (but do not eliminate) OLS bias.[47] Equations (130) to (133) provide a more realistic example, where, for ease of exposition, we again assume that the disturbance follows an AR(1) process:

$$y_t = \beta y_{t-1} + \gamma x_t + u_t \qquad \text{(equation to be estimated)} \tag{130}$$

where all variables are mean deviates and x_t is an exogenous variable independent of u_t.

$$u_t = \phi_1 u_{t-1} + \nu_t \qquad \text{(disturbance time-dependence process)} \tag{131}$$

$$\left.\begin{array}{l} -1 < \beta < 1 \\ -1 < \phi_1 < 1 \end{array}\right\} \qquad \text{(stationarity conditions)} \tag{132}$$

$$\left.\begin{array}{l} E(u_t) = E(\nu_t) = E(u_{t-\theta}\nu_t) \\ \quad\;\; = E(y_{t-\theta}\nu_t) = 0 \end{array}\right\} \text{for } \theta > 0 \text{ and all } t \tag{133}$$

Equation (130) might have included additional lagged $y_{t-\theta}$ and contemporaneous or lagged x_{kt} without fundamentally affecting the results that follow. Note that we assume x_t to be a genuine exogenous variable independent of u_t, but of course this is not true of y_{t-1}, which is necessarily linked to u_t via the serial dependence.

Performing operations analogous to Eqs. (123) to (125) generates the following expression for the systematic component of the disturbances:

$$\phi_1 u_{t-1} = \phi_1 y_{t-1} - \phi_1 \beta y_{t-2} - \phi_1 \gamma x_{t-1} \tag{134}$$

Combining Eqs. (130) and (131) of the original model and substituting (134) for $\phi_1 u_{t-1}$ allows the equation for y_t to be rewritten as

$$\begin{aligned} y_t &= \beta y_{t-1} + \gamma x_t + \phi_1 y_{t-1} \\ &\quad - \phi_1 \beta y_{t-2} - \phi_1 \gamma x_{t-1} + \nu_t \\ y_t &= (\beta + \phi_1) y_{t-1} + \gamma x_t - \phi_1 \beta y_{t-2} \\ &\quad - \phi_1 \gamma x_{t-1} + \nu_t \end{aligned} \tag{135}$$

Again it is apparent that the dynamic equation in (130), whose disturbance follows the mechanism in (131), may be transformed into a more tractable omitted-variables problem. Hence we have $(\beta + \phi_1)$ and γ as the "true" coefficients of the included variables y_{t-1} and x_t, and $-\phi_1\beta$ and $-\phi_1\gamma$ as the "true" parameters of the omitted variables y_{t-2} and x_{t-1}. It follows straightforwardly from Eq. (126) that OLS ap-

[47] Malinvaud (1961).

plied to the original equation does not consistently estimate β and γ, but rather:

$$\text{plim } \hat{\beta} = (\beta + \phi_1) + [-\phi_1\beta * b_{y_{t-2},\, y_{t-1}\cdot x_t} - \phi_1\gamma * b_{x_{t-1},\, y_{t-1}\cdot x_t}] \tag{136}$$

$$\text{plim } \hat{\gamma} = \gamma + [-\phi_1\beta * b_{y_{t-2},\, x_t\cdot y_{t-1}} - \phi_1\gamma * b_{x_{t-1},\, x_t\cdot y_{t-1}}] \tag{137}$$

where ϕ_1 plus the bracketed expression in (136) is the asymptotic bias of $\hat{\beta}$; the bracketed expression in (137) is the asymptotic bias of $\hat{\gamma}$; and $*$ denotes multiplication.

These results admit a variety of possibilities, but clearly the most common is an overestimate of β (the coefficient of y_{t-1}) and an underestimate of γ (the coefficient of x_t).[48] Therefore OLS estimation of dynamic models with autocorrelated disturbances typically produces upwardly biased coefficients for the lagged $y_{t-\theta}$ and downwardly biased (toward zero) coefficients for the exogenous x_{kt}. Since the lagged dependent variables tend to dominate the outcomes of such equations (partially) at the expense of exogenous influences, this leads to erroneous inferences about the importance of the dynamic terms. Finally the essentials of the results developed for simple autoregressive models concerning the asymptotic bias in $\hat{\phi}_1$ and $\hat{\rho}_1$ obtain here as well. OLS residuals cannot therefore be used to determine the structure or the magnitude of the disturbance time dependence.

So far we have shown that the parameters secured from OLS estimation of models with lagged endogenous variables and serially correlated disturbances do not accurately assess the true magnitude of causal influences. Moreover correlogram and related analyses are no longer feasible, and so pseudo-GLS estimation cannot be undertaken in the usual way. This brings us to an examination of estimation methods that are appropriate to the problem.

Estimation Methods for Dynamic Models with Serially Correlated Errors

If the structure as well as the parameters of the disturbance time-dependence process are known a priori, then the special estimation pro-

[48] Note, however, that β is inflated less here than in the pure autoregressive model of Eq. (119). If $\phi_1 < 0$, that is, if the disturbances are negatively autocorrelated, then the consequences are just the opposite: underestimation of β and overestimation of γ. However, negative serial correlation is rare except in certain types of transformed, distributed lag models. See Griliches (1961, 1967) and Wallis (1969). It should also be pointed out that these basic results apply to cross-section based models that incorporate lagged endogenous variables, although in such situations it is not possible to determine the structure of the disturbance autocorrelation.

cedures developed in what follows need not be considered—one may proceed directly with generalized least squares. It is obvious, however, that such knowledge is rarely, if ever, available in practice, and so we are left with two realistic situations that regularly arise in applied research.

First consider the case where the form but not the parameter values of the interdependence process is known.[49] Suppose, for example, we have a model of the sort in Eq. (138) whose disturbance can be assumed to follow a first-order autoregressive process.

$$y_t = \beta y_{t-1} + \gamma x_t + u_t \tag{138}$$

$$u_t = \phi_1 u_{t-1} + \nu_t \tag{139}$$

It was demonstrated earlier [see (135)] that the y_t in this model satisfy the revised equation:

$$\begin{aligned} y_t = {} & (\beta + \phi_1) y_{t-1} + \gamma x_t - \phi_1 \beta y_{t-2} \\ & - \phi_1 \gamma x_{t-1} + \nu_t \end{aligned} \tag{140}$$

Since the disturbance term ν_t of the revised model is serially uncorrelated, Eq. (140) has all the desirable properties for least-squares estimation. Two methods are available to derive the parameters (β,γ) and ϕ_1. The first involves simultaneous estimation of the coefficients (subject to the nonlinear constraints) via an appropriate nonlinear regression program.[50] Convergence to the true parameter values, or indeed to any values at all, is of course not guaranteed but is probable if the investigator provides good starting guesses for the iterative search procedure employed by most nonlinear algorithms. For example in Eq. (140) starting guesses in the range $0 < \phi_1 < 1$ and $0 < \beta < 1$ (that of γ would depend on the substantive context) would surely make convergence to the true parameter values more likely than relying entirely on the default options of the available nonlinear program. Although simultaneous, nonlinear, iterative, and related search techniques are con-

[49] Typically such information is based on prior theory. See, for example, Davis, Dempster, and Wildavsky's second model for congressional budgeting behavior (1966). Perhaps the best-known example of a theoretically defined process derives from distributed lag models in economics. The most common formulation is $y_t = \alpha \sum_{\theta=0}^{\infty} \lambda^\theta x_{t-\theta} + u_t$, where $|\lambda| < 1$ and u_t is white noise. Subtracting λy_{t-1} produces the estimating equation: $y_t = \alpha x_t + \lambda y_{t-1} + u_t - \lambda u_t$, where the disturbance now follows an MA(1) process. An enormous econometric literature deals with the specification and estimation of such models. The former is surveyed by Griliches (1967) and the latter is reviewed by Aigner (1971). A comprehensive treatment on both counts is provided by Dhrymes (1971). The estimation techniques developed by Box and Jenkins (1970), although largely ignored in economics (Aigner excepted), are also relevant here.

[50] For example, see Marquardt (1963).

ceptually complex and computationally expensive, the limited experimental evidence indicates that such methods perform very well even in small samples.[51]

A second way to estimate the coefficients of Eq. (140), and hence indirectly those of Eq. (138), is by direct application of OLS. Thus the estimating equation is

$$\begin{aligned} y_t = \alpha_1 y_{t-1} + \alpha_2 x_t + \alpha_3 y_{t-2} \\ + \alpha_4 x_{t-1} + \nu_t \end{aligned} \tag{141}$$

Since we know that $\alpha_1 = (\beta + \phi_1)$, $\alpha_2 = \gamma$, $\alpha_3 = -\phi_1\beta$, and $\alpha_4 = -\phi_1\gamma$, the parameter estimates of interest are readily deduced as follows:

$$\begin{aligned} \hat{\phi}_1 &= -(\hat{\alpha}_4/\hat{\alpha}_2) = -(-\hat{\phi}_1\hat{\gamma}/\hat{\gamma}) \\ \hat{\beta} &= (\hat{\alpha}_1 - \hat{\phi}_1) = (\beta + \phi_1) - \hat{\phi}_1 \\ \hat{\gamma} &= \hat{\alpha}_2 \end{aligned} \tag{142}$$

Alternatively we might retain only the estimate of the AR(1) parameter ϕ_1 from Eq. (142) to generate $\hat{\Omega}$, $\hat{\Omega}^{-1}$, or the transformation matrix $\hat{A}$, and then reestimate the original model via pseudo-GLS. This should yield somewhat sharper estimates of β and γ.

The principal limitation of the procedures developed above is the requirement of prior knowledge of the form of the interdependence. The desirable properties of GLS are not in general preserved if the time-dependence process is incorrectly specified—indeed, the investigator who proceeds on the basis of a misspecified process may do more damage than good. For example, a study by Robert Engle (1973) demonstrates that even in the low-order autoregressive world, OLS can be vastly superior to GLS when Ω is truncated. In particular, Engle shows in the context of a simple model with one exogenous regressor that if the true disturbance is AR(2) and the process is mistakenly assumed to be AR(1), OLS may dominate GLS.

Since prior knowledge about the structure of the disturbance time dependence is not typically available for models of interest to sociologists and political scientists, it is probably more useful to examine the case in which neither the form nor the coefficients of the interdependence process are known in advance. Consider, then, a model of the sort in Eq. (143), where the u_t are autocorrelated but the mechanism producing the serial dependence is not known a priori.

$$y_t = \beta y_{t-1} + \sum_k \gamma_k x_{kt} + u_t \tag{143}$$

What is required to obtain consistent and efficient estimates of the parameters in Eq. (143)? Consistency alone may be secured by using

[51] See Dhyrmes (1971, appendix), Morrison (1970), and Sargent (1968).

a two-stage instrumental variables (IV) procedure which parallels the two-stage least-squares solution to the simultaneous equations problem. In the first stage the systematic part of the lagged endogenous variable is created by regressing y_{t-1} on the exogenous, instrumental variables x_{kt} and x_{kt-1}. That is, we perform the regression:

$$y_{t-1} = \sum_k \alpha_k x_{kt} + \sum_k \delta_k x_{kt-1} + \eta_{t-1} \tag{144}$$

Hence the systematic part of y_{t-1}, designated as $\hat{y}_{t-1}$, is

$$\hat{y}_{t-1} = y_{t-1} - \hat{\eta}_{t-1} = \sum_k \hat{\alpha}_k x_{kt} + \sum_k \hat{\delta}_k x_{kt-1} \tag{145}$$

Consistent estimates of the parameters of the original model are now straightforwardly obtained by replacing y_{t-1} with its "purified" counterpart $\hat{y}_{t-1}$ in a second-stage OLS regression:[52]

$$y_t = \beta \hat{y}_{t-1} + \sum_k \gamma_k x_{kt} + \omega_t \tag{146}$$

where $\omega_t = u_t + \hat{\beta}\hat{\eta}_{t-1}$.

All variables are now asymptotically uncorrelated with the composite disturbance ω_t. This is assured because the current and lagged x_{kt} are uncorrelated in the probability limit with the u_t by assumption, and are uncorrelated in the sample with the $\hat{\eta}_{t-1}$ by construction. Since $\hat{y}_{t-1}$ is created from a linear combination of the x_{kt}, it too is asymptotically uncorrelated with the composite disturbance ω_t. Hence the two-stage IV method of Eqs. (145) and (146) is robust against all forms of serial dependence and renders consistent estimates of β and γ_k.[53] However, IV estimation may produce a considerable loss of precision, and although it is generally superior to OLS in large samples, the small-sample evidence has not been encouraging.[54]

[52] Equivalently one may proceed with classic IV estimation and use $\hat{y}_{t-1}$ as an instrument instead of actually replacing y_{t-1} with it. The estimating equation would then be $\hat{B} = (z'x)^{-1} z'y$, where $\hat{B} = (\hat{\beta}, \hat{\gamma}_k)$; $z = (\hat{y}_{t-1}, x_{kt})$; and $x = (y_{t-1}, x_{kt})$. See Liviatan (1963). These procedures are easily adapted to the general case of m lagged endogeneous variables as long as there are sufficient instrumental variables available—a condition known as the *identification problem* in the simultaneous equations literature. See Fisher (1966, 1970) and Hibbs (1973).

[53] Provided that $\hat{y}_{t-1}$ and x_{kt} are not connected by linear identities; that is, the current x_{kt} and the lagged x_{kt-1} must not be connected by nonstochastic, linear, first-order difference equations. Also, note that Eq. (146) does not yield appropriate estimates of R^2, σ_u, and so on. These are derived by using the consistent estimates of β and γ_k from Eq. (146) in conjunction with the original data and model to generate goodness of fit and error variance statistics. See Eq. (147).

[54] See Malinvaud (1961), Morrison (1970), Sargent (1968), and Wallis (1967).

It is therefore advantageous to utilize, but extend further, the IV procedure in order to secure more efficient pseudo-GLS estimates. This is undertaken as follows. First use the consistent estimates of β and γ_k from the second-stage IV regression of Eq. (146) in conjunction with the original data and model to form estimates of the original disturbances u_t. That is, calculate $\hat{u}_t$:

$$\hat{u}_t = y_t - \left[\hat{\beta} y_{t-1} + \sum_k \hat{\gamma}_k x_{kt}\right] \tag{147}$$

Unlike the $\hat{u}_t$ obtained from direct OLS analysis of the initial model, the residuals derived from (147) are deduced from consistent parameter estimates applied to the original data. Accordingly they may be subjected to autocorrelation-correlogram analysis in order to determine the structure [AR(p), MA(p)] and coefficients ϕ_p of the disturbance time-dependence process via the procedures described previously.[55]

It is now easy to secure consistent pseudo-GLS estimates of (β,γ_k) and $\mathrm{var}(\hat{\beta},\hat{\gamma}_k)$:

$$\hat{B}^* = (X'\hat{\Omega}^{-1}X)^{-1}X'\hat{\Omega}^{-1}Y \tag{148}$$

or alternatively

$$\begin{aligned} \hat{B}^* &= (X^{*\prime}X^*)^{-1}X^{*\prime}Y^* \\ \mathrm{var}(\hat{B}^*) &= \sigma_u^2(X'\hat{\Omega}^{-1}X)^{-1} \end{aligned} \tag{149}$$

or alternatively

$$\mathrm{var}(\hat{B}^*) = \sigma_u^2(X^{*\prime}X^*)^{-1}$$

where X denotes a matrix of right-hand side variables (Y_{t-1}, X_t); X^* denotes a matrix of transformed right-hand side variables (AY_{t-1}, AX_t); Y^* denotes a vector for the transformed dependent variable AY_t; and $\hat{B}^*$ denotes a vector of GLS estimates $(\hat{\beta}^*,\hat{\gamma}^*)$.

The existing Monte Carlo evidence, based on artificial samples of 50 observations, demonstrates that the IV-GLS method outperforms OLS and IV alone for models such as Eq. (143), whose disturbance is first-order autoregressive.[56] There is no reason to believe that IV-GLS performs less well in situations where the form of the process is not assumed in advance, although this surely deserves further investigation.[57] At this point a substantive example is clearly in order.

[55] If correlogram analysis indicates that a moving average process is operative, then nonlinear estimation of the ϕ_p is necessary when $p > 1$. The ϕ_p estimates should also be corrected for small-sample bias before undertaking GLS. See my discussion and footnotes 31–34 for the particulars.

[56] See Wallis (1967).

[57] Two points should be added here. First, although the IV-GLS method is based on a consistent estimate of Ω and therefore yields consistent parameter estimates, the latter are not asymptotically efficient (unlike the case considered

Dynamics of Pre-World War I Arms Races

As an example of the issues surrounding the estimation of dynamic equations with autocorrelated disturbances, in which both the structure and the coefficients of the time-dependence process must be deduced from sample data, we reevaluate the British arms expenditures equation from Choucri and North's research on the causes of World War I.[58] Choucri and North's work is an impressive effort to model the most important dynamic interrelationships among the six great powers (Great Britain, Italy, France, Germany, Austria-Hungary, and Russia) that ultimately led to the outbreak of World War I. Naturally arms expenditures play a central role in the model.

In their single-equation analyses, Choucri and North specify Great Britain's defense expenditures during the 1871–1914 period (GB-Mil$\$_t$) to be a linear function of:

1. Defense expenditures during the previous year (GB-Mil$\$_{t-1}$) —the dynamic term representing the independent influence of bureaucratic-organizational momentum.
2. The sum of the previous year's defense expenditures of non-allied, competitor nations in the six-power system (Adversary-Mil$\$_{t-1}$)—which incorporates the arms-race, adversary-stimulation effect.
3. A scaled variable that measures the peak intensity of hostile interactions over spheres of influence (Clash$_t$)—a term scored from 1 to 30 that includes such interactions as disputes over patterns of influence in client states and confrontations over control of colonial territory between Britain and the other great powers.
4. The size of Britain's colonial area, log transformed (ln Col-area$_t$)—which is designed to capture the defense imperative of territorial-colonial expansion.
5. A variable representing the multiplicative effect of population,

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previously where only exogenous variables appear on the right-hand side). Second, Maddala (1971) has shown analytically that if the disturbance is autoregressive it pays to iterate Eq. (148), which amounts to an iterative solution of the relevant maximum-likelihood equations. However, if the disturbance follows an MA process defined by the model in footnote 49, then Hannan's (1965) two-step GLS procedure [which uses $\hat{Y}_{t-1}$ instead of Y_{t-1} in the right-hand side of (148)] amounts to an iterative solution of the maximum-likelihood equations. What these asymptotic results imply for the small-sample properties of the estimators remains to be definitively resolved.

[58] See Choucri and North (1972a, 1972b). The data for this example were kindly supplied by Professor Choucri.

or internal demands, and iron and steel production, or industrial capabilities and resources (Pop * Capabilities$_t$)—a measure of the domestic pressure for external expansion that is hypothesized to be a joint function of population size and level of technological-industrial capabilities.

The Choucri-North model for British arms expenditure during the 44 years preceding the onset of World War I can therefore be expressed as follows:

$$\begin{aligned}\widehat{\text{GB-Mil\$}}_t = \beta_0 &+ \beta_1 \text{ GB-Mil\$}_{t-1} \\ &+ \beta_2 \text{ Adversary-Mil\$}_{t-1} + \beta_3 \text{ Clash}_t \\ &+ \beta_4 \ln \text{ Colarea}_t + \beta_5 \text{ Pop} * \text{Capabilities}_t \\ &+ u_t\end{aligned} \tag{150}$$

Table 3 reports the OLS estimates for Eq. (150). Unless great parts of previous sections are in error, these results should exaggerate the influence of the lagged endogenous variable GB-Mil\$$_{t-1}$, and understate the impact of most of the exogenous causal factors—assuming, of course, that the disturbances are autocorrelated.

Since neither the form nor the parameters of the disturbance time-dependence process are known, an IV-GLS procedure is appropriate. Thus we first create the systematic part of GB-Mil\$$_{t-1}$ to obtain consistent estimates of the regression coefficients, which can then be applied to the original data and model to yield estimates of the true disturbances. The relevant IV equation is

$$\begin{aligned}\text{GB-Mil\$}_{t-1} = \hat{\alpha}_0 &+ \hat{\alpha}_1 \text{ Adversary-Mil\$}_{t-1} + \hat{\alpha}_2 \text{ Adversary-Mil\$}_{t-2} \\ &+ \hat{\alpha}_3 \text{ Clash}_t + \hat{\alpha}_4 \text{ Clash}_{t-1} + \hat{\alpha}_5 \ln \text{ Colarea}_t \\ &+ \hat{\alpha}_6 \ln \text{ Colarea}_{t-1} + \hat{\alpha}_7 \text{ Pop} * \text{Capabilities}_t \\ &+ \hat{\alpha}_8 \text{ Pop} * \text{Capabilities}_{t-1}\end{aligned} \tag{151}$$

Consistent estimates of the regression coefficients in the original model are now secured by estimating the second-stage equation:

TABLE 3
OLS Estimates for Equation (150)

Independent Variable	Parameter Estimate	Standard Error	t-Statistic
Intercept	$-3.61E + 6$	$1.11E + 6$	-3.26
GB-Mil\$$_{t-1}$	0.55	0.10	5.73
Adversary-Mil\$$_{t-1}$	0.10	0.03	4.01
Clash$_t$	-915.03	959.07	-0.95
ln Colarea$_t$	$4.08E + 5$	$1.24E + 5$	3.28
Pop*Capabilities$_t$	$-2.28E - 4$	$7.17E - 5$	-3.18

$R^2 = 0.900$ Regression standard error $= 4.37E + 4$ $F = 68.6$; $DF = 5{,}38$

$$\begin{aligned}\text{GB-Mil\$}_t = {} & \beta_0 + \beta_1 \widehat{\text{GB-Mil\$}}_{t-1} \\ & + \beta_2 \text{ Adversary-Mil\$}_{t-1} + \beta_3 \text{ Clash}_t \\ & + \beta_4 \ln \text{Colarea}_t + \beta_5 \text{ Pop} * \text{Capabilities}_t \\ & + \omega_t\end{aligned} \tag{152}$$

Finally we use the coefficients of Eq. (152) in conjunction with the original data to form estimates of the disturbances u_t:

$$\begin{aligned}\hat{u}_t = {} & \text{GB-Mil\$}_t - [\hat{\beta}_0 + \hat{\beta}_1 \text{ GB-Mil\$}_{t-1} \\ & + \hat{\beta}_2 \text{ Adversary-Mil\$}_{t-1} + \hat{\beta}_3 \text{ Clash}_t \\ & + \hat{\beta}_4 \ln \text{Colarea}_t + \hat{\beta}_5 \text{ Pop} * \text{Capabilities}_t]\end{aligned} \tag{153}$$

The disturbance estimates of Eq. (153), unlike those of the OLS results in Table 1, can legitimately be analyzed to determine the structure and coefficients of the time-dependence process. Figure 6 depicts

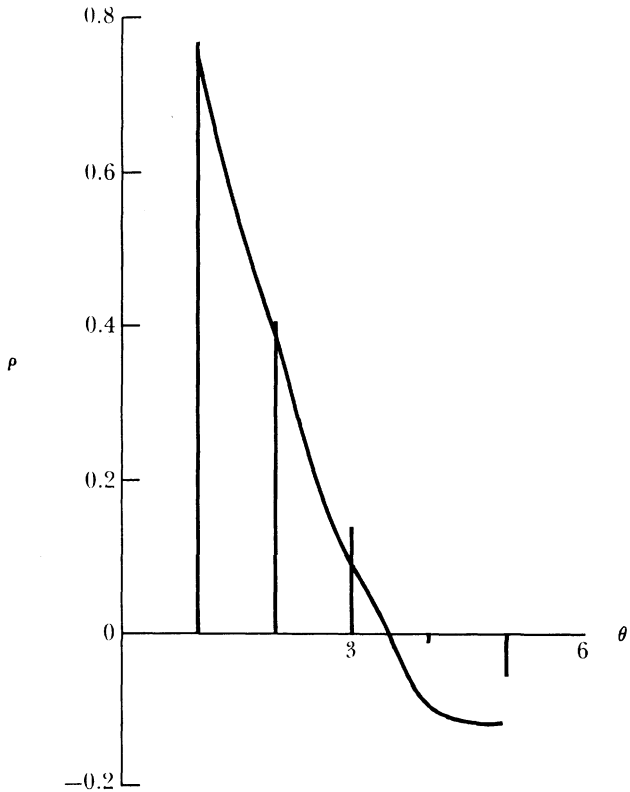

Figure 6. Empirical and theoretical correlograms for Equation (150). Curved line plots empirical autocorrelations; bar lines plot theoretical autocorrelations.

TABLE 4
GLS Estimates for Equation (150)

Independent Variable	Parameter Estimate	Standard Error	t-Statistic
Intercept	$-5.50E+6$	$1.74E+6$	-3.15
GB-Mil$\$_{t-1}$	0.31	0.14	2.29
Adversary-Mil$\$_{t-1}$	0.08	0.03	2.84
Clash$_t$	-274.34	709.48	-0.39
ln Colarea$_t$	$6.20E+5$	$1.94E+5$	3.19
Pop*Capabilities$_t$	$-2.51E-4$	$1.12E-4$	-2.24

$R^2 = 0.593^a$ Regression standard error $= 3.81E+4^a$ $F = 11.07^a$; $DF = 5{,}38$

[a] These GLS statistics should not be compared to the corresponding OLS statistics because they are based on the transformed disturbances AU. See the previous discussion, especially footnote 21.

the empirical (curved line) and theoretical (bar lines) correlograms of the best-fitting model—a second-order autoregressive process with coefficients $\hat{\phi}_1 = 1.0$ and $\hat{\phi}_2 = -0.362$.

It is now possible to undertake pseudo-GLS estimation of the initial Choucri-North model for pre-World War I British military expenditures. Table 4 displays the results, which contrast sharply with the biased and inconsistent OLS estimates in Table 3. As one would expect from the statistical theory developed in previous sections, the most important difference involves the coefficient of lagged military expenditures—it decreases in size by about 44 percent. This vitiates considerably Choucri and North's conclusion that the single most important factor affecting the current defense budget is the level of the budget at the previous time period. The bureaucratic-organizational or incrementalist effect is simply not predominant.[59] For example, the standardized regression coefficient or path coefficient of GB-Mil$\$_{t-1}$ (not reported here) exceeds that of all other variables in the OLS results but drops to third place in the GLS results.

Finally the coefficient estimates of the exogenous, independent variables also conform to statistical theory, being on the whole larger in the GLS outcomes than in the OLS results. Specifically the parameter of Adversary-Mil$\$_{t-1}$ is a bit smaller;[60] the parameter of Clash$_t$ remains

[59] I suspect that a great many incrementalist theories—which posit the previous year's expenditure or behavior as having a prevailing influence on the current year's outcome—would fare rather poorly when properly estimated.

[60] Although the military expenditure of adversary nations was specified in the early Choucri-North work as exogenous, and has been so treated in the replication here, it is clear from the arms race literature that Adversary-Mil$ is

insignificant; and the coefficients of ln Colarea_t and $\text{Pop} * \text{Capabilities}_t$ are significantly larger. Therefore it appears that OLS produced an inflated estimate of the causal influence of lagged expenditures while understating the impact of colonial expansion and internal population-resource pressure.[61]

CONCLUDING REMARKS

I have argued that to proceed with OLS regression in the presence of serially correlated disturbances can seriously impair the statistical estimation-causal inference process. At a minimum OLS produces inflated t, F, and goodness-of-fit statistics and can lead to spurious attribution of significance to independent variables and to exaggerated claims about the success of a model in explaining the phenomena being investigated. In large-scale studies, where a great many equations are estimated sequentially in order to develop final functional forms, initial errors of inference may be compounded—because if the process of model formulation and causal inference is cumulative, then so may be the ultimate impact of mistaken inferences made along the way.

The consequencies for estimation of dynamic models, which incorporate lagged endogenous variables on the right-hand side, are even more discouraging. Here consistency—the minimal property of any estimator—is not ensured. Hence neither the absolute nor the relative values of the causal parameters can be taken seriously.

Ideally the causes of autocorrelated disturbances should be included explicitly in an equation, thereby obviating the necessity of pursuing rather complicated estimation techniques designed to deal with the problem. However, this would require that the investigator identify the systematic errors of measurement, the serially correlated minor influences omitted from the model, and other subtle errors in functional form that can produce interdependent disturbances. This simply is not feasible for most models confronted in applied social research, and so we shall have to live with the difficulties of parameter estimation via alternatives to OLS regression.

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really endogenous. Hence the GLS results that give smaller coefficients and t-statistics for Adversary-Mil$\$_{t-1}$ as well as for GB-Mil$\$_{t-1}$ (and for that matter for Clash_t) are entirely compatible with the statistical theory developed above. Choucri and North's current reformulation of the model takes these points fully into account.

[61] Note, however, that the substantial impact of the Pop*Capabilities variable is opposite in sign to that anticipated by the Choucri-North theory in both the OLS and GLS results.

Finally, all the problems surveyed in this chapter apply to multi-equation models as well. Recursive causal systems pose no special difficulty in the sense that the techniques outlined earlier can be straightforwardly employed.[62] Simultaneous equations formulations, however, present additional complications that cannot be developed here without extending this essay far beyond its present length, and in any case have yet to be resolved completely.[63]

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[62] Recall that such systems require causal influences to flow hierarchically or unidirectionally (the matrix of endogenous-variable coefficients is triangular) and also require disturbances to be uncorrelated across equations (the cross-equation disturbance variance-covariance matrix must be diagonal). If a lagged endogenous variable appears in such a model and disturbances are autocorrelated, the latter assumption of recursive models must break down and the techniques presented in the last section become appropriate. Otherwise the procedures developed in earlier sections are servicable.

[63] See Amemiya (1966), Fair (1970), and Sargan (1961).

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